The Market-Perceived Natural Rate of Interest*

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Abstract

The natural rate of interest (r*) is a central reference point in gauging the stance of monetary policy, yet existing evidence on the real rate gap focuses primarily on model-based estimates adopted by the central bank, leaving market-based assessments largely unexplored. This paper estimates the natural rate of interest inferred from more than four decades of survey forecast data for the United States, using a dispersion-weighted state-space model that down-weights quarters of high forecaster disagreement rather than excluding periods of uncertainty such as the pandemic. We document a secular decline in perceptions of the natural rate and identify persistent episodes of misalignment between these beliefs and the Federal Reserve's benchmark rate estimated using realised data. We further show that macroeconomic fundamentals shape revisions in perceptions of equilibria, and that misalignments matter for financial markets, insofar as underestimation of the natural rate loosens financial conditions, raises equity valuations and expands credit, whereas overestimating r* has broadly the opposite effect.

Keywords: natural rate of interest, monetary policy, survey forecasts, perceptions

JEL classification: C32, E43, E47, E52

*We would like to thank participants at the Annual Conference of the Royal Economic Society, the Annual Conference of the Money Macro and Finance Society, the International Conference on Applied Theory, Macro and Empirical Finance, and the Annual Meeting of the European Economics and Finance Society, for their comments and suggestions.

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1 Introduction

Whether a given policy rate is restrictive or accommodative depends on where it sits in relation to the natural rate of interest (r*), the real interest rate consistent with stable inflation and output at potential. In practice, however, monetary policy works through the expectations of private agents, who form their own view of equilibrium based on their individual perceptions and forecasts of the economy. If the stance of policy is assessed relative to an equilibrium that agents misjudge, then the transmission of monetary policy depends not only on where the policy rate sits but on where agents believe it does relative to the natural rate of interest. That belief regarding the natural rate is neither observed nor reported over any long span of history.

In this paper, we construct a measure of these perceptions. Using professional survey forecasts for the United States from 1981:3 to 2025:4, we estimate the natural rate implied by private-sector expectations, hereafter the market-perceived natural rate, and compare it with a benchmark natural rate estimated from realised data using the same underlying semi-structural model. Three central results emerge. First, the perceived natural rate has declined by some 250 basis points, consistent with the secular decline in r* itself. Second, it has nonetheless departed from the benchmark rate for sustained periods, with forecasters persistently underestimating the natural rate for roughly two decades before the financial crisis and overestimating it for more than five years thereafter. Third, these misalignments have systematic effects on financial markets, with underestimation loosening financial conditions and overestimation tightening them.

Our analysis departs from the literature's extensive focus on documenting and explaining the decline in the natural rate of interest itself. Most prominently within this literature, Laubach and Williams (2003) and Holston et al. (2017; 2023) estimate r* from realised data and find that it has declined persistently in the United States since the 1960s, a trend commonly attributed to factors such as an ageing population (Carvalho et al., 2016; Eggertsson et al., 2019; Gagnon et al., 2021), declining productivity (Rachel and Smith, 2017), an increasing scarcity of safe assets (Caballero and Farhi, 2018; Del Negro et al., 2017), and shifts in fiscal and institutional arrangements (Rachel and Summers, 2019).[1] Much less is known about how the natural rate is perceived by those whose expectations it is supposed to anchor, and whether those perceptions align with it.

[1] Refer to Del Negro et al. (2017) for an established structural approach based on a New Keynesian model, in which the natural rate is defined as the real interest rate that would prevail under flexible prices, and Lubik and Matthes (2015) for an established time-series approach based on a time-varying-parameter VAR, in which the natural rate is identified from the model's long-horizon forecast of the real interest rate.

In this regard, we make three primary contributions. The first is one of measurement. We estimate a modified version of the semi-structural model of Holston et al. (2017; 2023) by maximum likelihood and the Kalman filter on forecasts of output, inflation and the short-term interest rate from the Survey of Professional Forecasters. Our modification involves scaling the covariance of the measurement errors according to the cross-sectional dispersion of the underlying survey itself. Quarters in which forecasters disagree sharply are thereby down-weighted, so that extreme periods of uncertainty such as during the pandemic can be retained rather than discarded. We show that this approach matters, since excluding the pandemic outright produces estimates that are sensitive to where the exclusion window is drawn. We find that the resulting time series follows a similar secular decline documented for the natural rate of interest itself, and that this decline is also shared across individual respondents rather than confined to the consensus.

Our second contribution is a measure of misalignment between these perceptions and the natural rate of interest itself, and an account of why it persists. We find that forecasters underestimate the natural rate of interest for roughly two decades prior to the financial crisis, with the gap widening through the productivity acceleration of the late 1990s. They then persistently overestimate r* until 2015, after which the two measures remain closely aligned for the remainder of the sample. These misalignments coincide with periods over which the natural rate was itself shifting, and in each case perceptions move in the same direction but with an attenuated response, tracking the secular path of the natural rate of interest rather than its shorter-run movements. Using local projections and vector autoregressions, we show that macroeconomic fundamentals drive revisions in the perceived natural rate, and that the response to innovations in the natural rate itself, the largest of these, builds over several years rather than on impact. Perceptions converge on the benchmark slowly and incompletely. This gives empirical content to models with incomplete information in which misperceptions of r* are large and persistent (e.g., Rungcharoenkitkul and Winkler, 2022), and implies that forecasters have historically misread the stance of monetary policy for years at a time. In particular, underestimating the equilibrium rate means judging policy to be tighter than it is, while overestimating it means judging policy to be looser than it is.

Third, we trace the consequences of these misalignments for financial markets. Whereas perceptions of the monetary policy reaction function have been shown to matter for asset prices (Bauer et al. (2024), the financial consequences of misperceiving the natural rate are less understood. To address this, we use local projection techniques to understand the impact that departures in beliefs

from the equilibrium real interest rate have across key financial indicators. Separating episodes of under- and overestimation, we find that underestimation of the natural rate of interest is followed by a loosening of financial conditions, a sustained rise in equity valuations, and an expansion of credit relative to output, with responses to overestimation carrying the opposite sign. Understood together, these findings indicate that misalignment between the beliefs of the private sector and the equilibrium interest rate is a systematic feature of the transmission mechanism rather than random noise, with implications for central bank communication and financial stability.

We subject these findings to a range of robustness checks. Re-estimating the model separately for each respondent, either holding parameters at their baseline values or re-estimating them, yields individual paths that trace the same secular decline, with dispersion concentrated in the tails once parameters are relaxed. Our estimates are largely insensitive to the window used to normalise the dispersion weighting, and the drivers of perceptions are broadly similar whether estimated by local projections or a vector autoregression. The perceived natural rate also aligns closely with longer-run policy rate expectations reported directly in central bank surveys in the years before 2017 and again towards the end of the sample, while its long-run decline is mirrored, if more sharply, in the trend component of real forward rates.

The rest of this paper is organized as follows. Section 2 briefly reviews the existing literature regarding market perceptions of monetary policy and the natural rate. Section 3 discusses particulars concerning the survey forecast data used in our estimation. Section 4 outlines the framework, implementation, and specification used to estimate the perceived natural rate of interest. Section 5 presents and discusses the main findings of our analysis. Finally, Section 6 concludes.

2 Market Perceptions

Existing research on perceptions and monetary policy has largely concerned market expectations about the future path of policy and the implications of incomplete information for its transmission (e.g. Blinder et al., 2008; Eusepi and Preston, 2010; Taylor and Williams, 2010; Cogley et al., 2015). A prominent strand of this research estimates the policy rule that market participants perceive the central bank to obey. Hamilton et al. (2011) estimate this 'market-perceived policy rule' from macroeconomic news, defined as the difference between released data and the value expected beforehand by the market, and find that the perceived response to output declined prior to the

Great Recession, while the response to inflation became more gradual but larger in the long run. Subsequent work has incorporated survey and individual forecasts to estimate constant-parameter monetary policy rules (Carvalho and Nechio, 2014; Andrade et al., 2016; Kim and Pruitt, 2017; Jia et al., 2023). More recently, Bauer et al. (2024) extend this to a time-varying framework using panel data on professional forecasts, showing that the perceived dependence of the policy rate on economic conditions changes substantially over time, that forecasters update their perceptions in response to high-frequency monetary policy surprises, and that these perceptions affect the sensitivity of interest rates to macroeconomic news, term premia in long-term bonds and the response of equity prices to policy surprises.

This literature has focused primarily on perceived reaction function coefficients, namely how strongly markets believe policy will respond to the inflation and output gap. Yet a given policy rate is only accommodative or restrictive relative to where the natural rate is judged to lie, so that agents who share the central bank's reaction function but not its view of equilibrium may nonetheless misread the stance of policy. Nor need such misperceptions be short-lived. Rungcharoenkitkul and Winkler (2022) develop a canonical New Keynesian model with incomplete information in which a feedback loop arises between the central bank and the private sector, each underestimating the impact of its own decisions on the other's understanding of r*. The resulting recursion can produce large and persistent shifts in the perceived natural rate disconnected from fundamentals, and may account for much of the decline in real interest rates since 2008.

A related strand recovers perceptions of the natural rate of interest from asset prices. Laubach and Williams (2016) briefly examine real forward rates as a measure of market participants' perceptions of the natural rate, applying univariate time-series methods, but conclude that term premia render such rates an unreliable guide to expected future interest rates, which motivates their multivariate approach adapted in this paper. More recently, Ahonon and Roussellet (2026) introduce a term structure model in which investors do not observe the long-run trends in interest rates, inflation and growth, but must infer them from incomplete information. Bond yields subsequently reflect perceived rather than true trends, and the resulting estimate of the perceived natural rate is relatively stable over six decades, though subject to wide uncertainty.

Survey-based approaches avoid the need to model risk premia, but rely on direct questions about the longer-run policy rate that have only recently been asked, such that available samples remain short. Benigno et al. (2024) include, among the range of measures they review in assess-

ing the natural rate after the pandemic, a perceived natural rate defined as the difference between the long-run expected policy rate and expected inflation, using median responses to the Survey of Primary Dealers. Results suggest perceptions may have risen over the pandemic, but that these beliefs are surrounded by high levels of uncertainty. Hartley (2026) constructs model-free measures from several central bank surveys of market participants, finding that survey-based estimates rose after the pandemic whilst structural estimates exhibit mixed dynamics, and that the former closely track the Federal Reserve's longer-run projections of the federal funds rate.

Cho and Williams (2025) compare a measure derived from the Blue Chip survey, alongside term structure model-based measures and real-time estimates. Their findings suggest that survey-based measures declined more gradually after the financial crisis and that the low natural rate era continues to prevail. Survey expectations have also featured in the estimation of r* itself. Zaman (2025) develops a semi-structural model that jointly estimates the long-run equilibrium levels of output, unemployment, the real interest rate, productivity and inflation, linking these to long-run survey expectations and allowing for time-variation and stochastic volatility. The survey data help to discipline the estimates, and prove crucial to estimation since the pandemic.

We build on this literature by recovering the perceived natural rate from survey forecast data, which complements existing survey- and market-based measures of the natural rate without relying on direct responses from market participants. To achieve this result, we adapt the semi-structural framework underlying estimates of the natural rate of interest maintained by the Federal Reserve Bank of New York, so that market perceptions are measured on the same foundation as a widely used monetary policy benchmark. We discipline the estimation with a measure of forecaster disagreement provided by the Survey of Professional Forecasters, down-weighting quarters in which forecasters disagree sharply. Whereas periods of extreme uncertainty such as the coronavirus pandemic have often been excluded from estimation, this modification allows them to be retained and weighted according to their informational content.

Additionally, we introduce an explicit measure of misalignment between perceptions of the natural rate and the natural rate itself. Whilst existing research has focused primarily on estimates of the equilibrium interest rate and its role in the transmission mechanism of monetary policy, deviations of private-sector beliefs from the natural rate of interest have received comparatively little attention. Such deviations are equally as important to the assessment of the policy stance, since an agent who misjudges the natural rate may misread whether policy is restrictive or accommodative.

We therefore document the magnitude and persistence of this gap and further examine the extent to which macroeconomic fundamentals drive these perceptions and what misalignments imply for central measures of financial conditions.

3 Survey Data

We begin by describing the details of our survey data set before outlining the techniques we use to estimate the market-perceived natural rate. We collect survey-based forecast data on the nominal short-term (three-month) interest rate, real gross domestic product, and consumer price inflation. We source these forecasts from the Survey of Professional Forecasters (SPF), published by the Federal Reserve Bank of Philadelphia, which is the longest continuously running quarterly survey of macroeconomic expectations in the United States and reports forecasts for each of the three variables entering our measurement equations at a consistent horizon. Our total sample runs from 1981:3 to 2025:4, with our estimation commencing in 1982:3 post initialisation.

The SPF is conducted quarterly, with each panellist reporting forecasts for horizons ranging from the current quarter to four quarters ahead. We employ the one-quarter-ahead forecast of each variable throughout, which matches the timing convention of the measurement equations set out in our model and avoids the horizon-smoothing conversions that would otherwise be required to align forecasts reported at different maturities. Since each respondent reports a nowcast alongside the one-quarter-ahead forecast within the same submission, we construct our measure of forecast real output by chaining the within-vintage growth rate implied by these two entries.[2] Forecasts of inflation and of the three-month rate are used as reported.[3]

We also collect the individual responses underlying each survey round, from which we construct two further objects. The first is the cross-sectional dispersion of forecasts in each quarter, measured by the interquartile range of the one-quarter-ahead responses for output growth and inflation, which enters the estimation through the dispersion weighting factor κ_t outlined in Section 4. The second is the panel of respondent-level forecasts itself, which allows us to recover a distribution of perceived natural rates across forecasters rather than a single consensus path, and so to assess whether our estimates are representative of the panel or an artefact of aggregation.

[2] Taking the log difference between one-quarter-ahead forecasts and the nowcast of real output within each vintage yields a growth rate that is internally consistent, since both entries are reported against the same underlying base year. Chaining these rates insulates the resulting level series from the periodic rebasing of the national accounts.

[3] Individual time-series for the one-quarter-ahead survey forecasts are presented in Appendix B.

Key summary statistics are presented in Table 1. Forecast CPI inflation averages approximately 2.8% over the sample and forecasts of the three-month rate 3.6%, with standard deviations of 1.4 and 2.8 percentage points respectively, reflecting the considerable variation in views concerning short-term nominal interest rates over a period spanning the Volcker disinflation, two decades at or near the effective lower bound, and the post-pandemic inflation surge. Forecast real output growth averages 2.2%, but its distribution is strongly left-skewed, with a minimum of -38.2% and a maximum of 17.6% corresponding to the contraction and rebound of 2020.

3.1 Extreme Tail Events

Our sample includes the coronavirus pandemic, a period that we would otherwise have reason to exclude given its unique impact on the macroeconomy. The pandemic was an extreme-tailed event that generated sharp variation in output growth, which violates the assumption of the Kalman filter that stochastic innovations are Gaussian in distribution. Furthermore, the sequence of restrictions imposed over the pandemic is inconsistent with the assumption that transitory shocks in the Phillips curve are serially uncorrelated. These problems within the measurement equations are at least as acute in forecast data as in realised data, since expectations of output growth responded sharply to the unforeseen imposition and relaxation of restrictions, and the uncertainty surrounding estimates of the natural rate of interest over this period is well documented in the existing literature (see Baker et al., 2023; Benigno et al., 2024).

In their well established model, Holston et al. (2023) address this issue along two dimensions, introducing time-varying volatility in the innovations to each measurement equation over the pandemic period, which suppresses the influence of outliers on the maximum likelihood estimates, and adding a persistent but transitory supply shock to the output gap to capture the effect of sequential lockdowns. The second of these adjustments is unavailable to us, insofar as controlling for the variation in forecast output induced by lockdowns would require forecasts of the restrictions themselves. The first, however, has a natural analogue in survey data. Where Holston et al. (2023) estimate the unrestricted variance scaling parameter over a fixed window, cross-sectional dispersion provides a direct and observable measure of how reliably the consensus forecast summarises a common view in any given quarter. In the following section, we outline our approach to exploiting this time-varying feature, allowing us to address the pandemic, as well as other periods of sharp uncertainty, rather than discarding them.

4 The Model

4.1 Framework

This paper adapts the Holston et al. (2017) framework to estimate implied forecasts of the natural rate $E_t r^*_{t+h}$ or $r^*_{t+h|t}$, where subscript $t+h$ represents the horizon of the forecast at t; we use this notation interchangeably. Given our focus is on the market-perceived natural rate, we preserve the model structure but use survey forecast data for each variable. In particular, we outline the system:

$$\tilde{y}_{t+h|t} = \hat{\alpha}_y(L)\tilde{y}_{t+h|t} + \hat{\alpha}_r(L)\tilde{r}_{t+h|t} + \varepsilon_{\tilde{y},t} \tag{1}$$

$$\pi_{t+h|t} = \hat{\beta}_\pi(L)\pi_{t+h|t} + \hat{\beta}_y(L)\tilde{y}_{t+h|t} + \varepsilon_{\pi,t} \tag{2}$$

$$r^*_{t+h|t} = g_{t+h|t} + z_{t+h|t} \tag{3}$$

$$y^*_{t+h|t} = \phi_{y*}(L)\, y^*_{t+h|t} + \phi_{y^*_g}(L)\, g_{t+h|t} + \varepsilon_{y*,t} \tag{4}$$

$$g_{t+h|t} = \phi_g(L)\, g_{t+h|t} + \varepsilon_{g,t} \tag{5}$$

$$z_{t+h|t} = \phi_z(L)\, z_{t+h|t} + \varepsilon_{z,t} \tag{6}$$

where (1)-(2) are the measurement equations, which are reduced forms of the Investment-Savings (IS) curve and Phillips Curve (PC) that permit shocks to the forecast output gap $\tilde{y}_{t+h|t}$ and forecast inflation $\pi_{t+h|t}$; (3) is the law of motion for the perceived natural rate $r^*_{t+h|t}$ that is a function of the forecast trend growth rate $g_{t+h|t}$ and the unobserved component $z_{t+h|t}$; and (4)-(6) are transition equations of the state-space system.[4]

The forecast output gap is defined as $\tilde{y}_{t+h|t} = y_{t+h|t} - y^*_{t+h|t}$, the deviation of forecast log output from the forecast potential output. The forecast real interest rate gap is likewise defined as $\tilde{r}_{t+h|t} = r_{t+h|t} - r^*_{t+h|t}$, the deviation of the forecast real rate from the perceived natural rate, where the former is the forecast nominal three-month rate less forecast inflation. Neither potential output nor the natural rate is reported by survey respondents. Both are treated as latent states within the system, recovered jointly within the filter as the paths most consistent with forecast output and inflation behaving in accordance with the dynamic Investment Savings equation and the Phillips

[4] The law of motion follows directly from the Ramsey growth model, contingent on representative households with a CES utility function and a constant relative risk aversion. In steady state, r-star is a function of the growth rate of per capita consumption and the rate of time preference. We assume forecasts of the natural rate are driven by similar forces.

curve. We do not construct forecast potential output prior to estimation, for instance by applying a statistical filter to forecast output, since doing so would then impose a trend-cycle decomposition independent of the model and prejudge the object of interest. Potential output is instead identified through the structure of the state-space system, where the standard Hodrick-Prescott filter is used only to initialise the state vector.

The state-space system identifies these unobservables through two core channels. The dynamic Investment Savings equation relates the forecast output gap to the gap between the forecast real rate and the perceived natural rate, so that $r^*_{t+h|t}$ is the real interest rate at which forecast output would neither rise above nor fall below its potential level. The Phillips curve links forecast inflation to the forecast output gap, anchoring potential output as the level consistent with stable forecast inflation. The perceived natural rate is then the sum of the trend growth rate of forecast potential output and a residual component capturing other determinants of the equilibrium real rate, each of which is a random walk, so that shocks to either have permanent effects on its level.

4.2 Adjustment

The baseline specification treats the measurement errors as homoskedastic, with a fixed covariance $\mathbf{R}$. In our setting, however, the observables are survey forecasts, such that in the presence of sharp disagreement the consensus is a noisier signal of the perceived equilibrium than in quarters where forecasters agree. We therefore allow the measurement-error covariance to vary over time, scaling it by a dispersion weighting factor $\kappa_t \geqslant 1$ constructed from the cross-sectional dispersion of the Survey of Professional Forecasters. We outline this adjustment in the notation of Hamilton (1994).

Formally, the observation equation in the state-space system is:

$$\mathbf{y}_t = \mathbf{A}'\mathbf{x}_t + \mathbf{H}'\boldsymbol{\xi}_t + \mathbf{w}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \tag{7}$$

where $\mathbf{y}_t$ captures the observables, $\mathbf{x}_t$ the predetermined variables and $\boldsymbol{\xi}_t$ the unobserved states. $\mathbf{A}$ and $\mathbf{H}$ are the corresponding coefficient matrices, and $\mathbf{w}_t$ collects the zero-mean and normally distributed measurement errors, with constant covariance $\mathbf{R}$.

We replace this measurement-error covariance with the time-varying covariance:

$$\mathbf{R}_t = \kappa_t^2 \, \mathbf{R}, \qquad \kappa_t \geqslant 1 \tag{8}$$

Such that in any quarter t, the innovation covariance in the Kalman filter becomes:

$$\mathbf{S}_t = \mathbf{H}' \mathbf{P}_{t|t-1} \mathbf{H} + \kappa_t^2 \mathbf{R} \tag{9}$$

where $\mathbf{H}$ is the measurement matrix and $\mathbf{P}_{t|t-1}$ the one-step-ahead state covariance. Since the Kalman gain $\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}\,\mathbf{S}_t^{-1}$ is decreasing in $\mathbf{S}_t$, a value $\kappa_t > 1$ inflates the observation variance, shrinks the gain, and thus causes the filter to update less on quarter t's survey forecasts. When $\kappa_t = 1$ the system collapses exactly to the baseline specification.

To construct κ_t, let D_t^y and D_t^π denote the cross-sectional interquartile ranges of the one-quarter-ahead forecasts of output growth and inflation respectively. We form a single scalar per quarter as the geometric mean of the two normalised dispersions:

$$\tilde{\kappa}_t = \sqrt{\frac{D_t^y}{\bar{D}^y} \cdot \frac{D_t^\pi}{\bar{D}^\pi}} \tag{10}$$

where $\bar{D}^y$ and $\bar{D}^\pi$ are the pre-2020 sample means of the dispersion series found in the SPF database, so that $\tilde{\kappa}_t$ is unit-normalised over the pre-pandemic period.[5]

We then impose the floor:

$$\kappa_t = \max\big(\tilde{\kappa}_t,\, 1\big) \tag{11}$$

following Holston et al. (2023). The restriction $\kappa_t \geqslant 1$ ensures that the adjustment can only inflate the observation variance, down-weighting high-disagreement quarters, and can never reduce it below the baseline, which would amount to placing above-baseline weight on low-disagreement quarters. The weighting is therefore unidirectional, damping outliers without allowing the estimation to over-trust quarters of forecaster consensus. This asymmetry reflects the assumption that high disagreement is informative about the reliability of the consensus, whereas low disagreement suggests forecasters have converged on a common view rather than its accuracy.

The transition dynamics and the deterministic structure of the measurement equations remain unchanged, except for the covariance of the measurement innovations. The key difference from the baseline model is that the measurement-error vector $\boldsymbol{\varepsilon}_t^m = (\varepsilon_{\tilde{y},t},\, \varepsilon_{\pi,t})'$ is now heteroskedastic, with a time-varying covariance governed by κ_t as follows:

[5] Our estimates are largely insensitive to the normalisation window. Changing the window rescales $\tilde{\kappa}_t$ proportionally, altering which quarters are down-weighted because of the floor, but the windows considered yield similar scale factors and quarters with large κ_t are already close to fully down-weighted, so further rescaling has little effect on the filtered estimates. We report estimates under alternative windows in Figure D.1, which show only modest shifts.

$$\boldsymbol{\varepsilon}_t^m \sim \mathcal{N}\big(\mathbf{0},\, \kappa_t^2\, \mathbf{R}\big), \qquad \mathbf{R} = \begin{bmatrix} \sigma_{\tilde{y}}^2 & 0 \\ 0 & \sigma_{\pi}^2 \end{bmatrix} \tag{12}$$

4.3 Specification

Our specification of the measurement equations of the state-space model is therefore analogous to Holston et al. (2017) in that we impose an analogous lag structure in the dynamic Investment Savings (IS) equation and Phillips Curve (PC). This decision is purely to facilitate direct comparability between the implicitly-perceived natural rate and the benchmark rate.[6]

In particular, the measurement equations are specified as:

$$\tilde{y}_{t+1|t} = \sum_{j=1}^{2} \hat{\alpha}_{y,j} \tilde{y}_{t-j+1|t-j} + \frac{\hat{\alpha}_r}{2} \sum_{j=1}^{2} \tilde{r}_{t-j+1|t-j} + \varepsilon_{\tilde{y},t} \tag{13}$$

$$\pi_{t+1|t} = \hat{\beta}_{\pi} \pi_{t|t-1} + \frac{1-\hat{\beta}_{\pi}}{3} \sum_{j=2}^{4} \pi_{t-j+1|t-j} + \hat{\beta}_y \tilde{y}_{t|t-1} + \varepsilon_{\pi,t} \tag{14}$$

where we incorporate two lags of the forecast output gap within the dynamic Investment Savings equation (13), in addition to the average of the first two lags of the forecast real rate gap. Within the Phillips Curve (14), we include the first and the average of the second-to-fourth lag of forecast inflation, in addition to the first lag of the forecast output gap.

The law of motion for the forecast natural rate is specified as:

$$r^*_{t+1|t} = g_{t+1|t} + z_{t+1|t} \tag{15}$$

The remaining transition equations are specified as:

$$y^*_{t+1|t} = y^*_{t|t-1} + g_{t|t-1} + \varepsilon_{y^*,t}, \tag{16}$$

$$g_{t+1|t} = g_{t|t-1} + \varepsilon_{g,t}, \tag{17}$$

$$z_{t+1|t} = z_{t|t-1} + \varepsilon_{z,t}, \tag{18}$$

[6] Lag-selection criteria provide some support for this structure. The lag length in the IS equation and the averaging restriction in the Phillips curve are largely consistent with the survey data, albeit the restriction that the coefficients on lagged inflation sum to one is mixed. We retain the restriction for comparability with the benchmark natural rate.

4.4 Implementation

Our model is estimated by the three-stage procedure of Holston et al. (2017; 2023), in the state-space notation of Hamilton (1994). Each stage of the model is estimated by maximum likelihood using the prediction-error decomposition with the κ_t-scaled innovation covariance (9),

$$\log L(\boldsymbol{\theta}) = -\frac{1}{2}\sum_{t=1}^{T}\Big[\, n\log(2\pi) + \log\big|\mathbf{S}_t(\boldsymbol{\theta})\big| + \mathbf{u}_t(\boldsymbol{\theta})'\,\mathbf{S}_t(\boldsymbol{\theta})^{-1}\,\mathbf{u}_t(\boldsymbol{\theta})\,\Big], \tag{19}$$

with $n = 2$, so that quarters of high dispersion enter the fit term $\mathbf{u}_t'\mathbf{S}_t^{-1}\mathbf{u}_t$ with reduced weight. Stage I recovers the signal-to-noise ratio $\lambda_g = \sigma_g/\sigma_{y^*}$, Stage II recovers $\lambda_z = |\hat{\alpha}_r|\sigma_z/\sigma_{\tilde{y}}$, and Stage III estimates the full system conditional on (λ_g, λ_z).

The dispersion weighting enters the two median-unbiased estimators asymmetrically. In Stage II it enters explicitly, since the structural-break regression is estimated by weighted least squares with weights $w_t = 1/\kappa_t^2$, so that high-dispersion quarters are discounted in the determination of λ_z exactly as they are in the filter. In Stage I the break statistic is computed upon the smoothed potential-output path, which is itself weighted, where the break regressions are unweighted, so that κ_t influences λ_g indirectly. Setting $\kappa_t = 1$ for all t reduces the procedure exactly to the unadjusted estimation, so that the adjustment is a strict generalisation of the baseline model, nested at $\kappa_t \equiv 1$. The complete stage-by-stage specification is reported in Appendix A.

5 Results

5.1 Parameter Estimates

The estimated dispersion weighting factor κ_t is presented in Figure 1. Values above unity denote quarters in which forecaster disagreement exceeds its pre-pandemic average, and in which the corresponding observation is down-weighted in the filter. The adjustment is inactive in well over half of the sample, so that for much of the period the model reduces to its unadjusted counterpart. Where the adjustment binds, it does so in known periods of volatility. For instance, disagreement is elevated throughout the first decade of the sample, during which inflation was relatively unstable. It is subsequently bounded at the floor during the 1990s, such that the Great Moderation is estimated with minimal weighting. A further episode accompanies the Great Recession of 2008, where κ_t approaches two, inflating the observation variance by a factor of four.

The pandemic produces by far the largest shift in the dispersion weighting, with κ_t reaching approximately 4.5 in 2020:2, inflating the observation variance in that quarter by a factor of more than twenty. Values remain above unity through the subsequent inflation surge before returning once again to the floor by 2024. The adjustment does not therefore rescale the sample uniformly; it down-weights those quarters in which the consensus forecast is least likely to represent a common view, and leaves the remainder of the sample unaltered.

We summarise the core parameter estimates of the model in Table 2. We employ the benchmark Holston et al. (2023) model to estimate the natural rate of interest (r*) using realised time series on real gross domestic product, consumer price inflation, and short-run (three-month) interest rates, which we also source exclusively from the Federal Reserve Bank of St. Louis (FRED) database.[7] Parameter estimates from this model are presented simultaneously in Table 2.

The median-unbiased estimate of λ_g is approximately half as large in the forecast model, while λ_z is essentially unchanged. Combined with the innovation variances, trend growth accounts for about one third of the variance of the market-perceived natural rate, against just over half for the benchmark. Time variation in the perceived rate therefore derives relatively more from the residual component z than from revisions to expected trend growth. The IS-curve slope α_r is close to the imposed bound in both models, and the Phillips-curve slope β_y is closer to its lower bound for the forecast model, thus neither is tightly identified, consistent with the difficulty documented in the existing literature. The forecast output gap is nonetheless persistent, as indicated by the cumulative magnitude of $\hat{\alpha}_y$ on its own lags. Finally, it is also worth noting that the innovation variances are broadly comparable to those obtained from realised data.

Where the two sets of estimates differ notably is in precision. Average standard errors of the market-perceived natural rate are relatively larger than those of its benchmark model counterpart, which is to be expected given the volatility of expectations concerning the future path of output, inflation and interest rates. Such imprecision is a well known artefact of the literature due to both parameter and filter uncertainty, and our estimates are no exception. As there is no clear resolution to this particular issue, we approach the level of the market-perceived natural rate with caution and place greater weight on its movements, and on its difference from the benchmark rate.

[7] We re-estimate the model rather than source published estimates directly, since the latter rely on core PCE inflation and the federal funds rate, whereas the survey forecasts are of CPI inflation and the three-month Treasury bill rate, which ensures both series are based on the same measures. This common implementation also means that the distance between the two series reflects the inputs alone, given an identical sample, parameter constraints and initialisation.

5.2 Market-Perceived Natural Rates

Estimates of the perceived natural rate as implied from survey forecast data are presented in Figure 2. The series peaks in the mid-1980s at close to 4%, falling to around 2% by the early 1990s, and varies marginally over the following fifteen years. It then continues to decline through the financial crisis, reaching its lowest point in 2015 and remaining below its pre-crisis level for the remainder of the sample. Our estimates suggest that the implied equilibrium interest rate has fallen by some 250 basis points from its mid-1980s peak.

We present benchmark estimates of the natural rate of interest using realised data alongside our results in Figure 3.[8] These findings reveal how the long-run trend of the implied natural rate closely resembles that of the natural rate of interest itself. Such persistent pessimism concerning the future level of r* is in keeping with secular declines in the underlying natural rate of interest itself, attributed in the literature to long-run shifts in productivity, demography, risk preferences and public policy. That market expectations of the natural rate have moved in the same direction and by a comparable magnitude suggests that these underlying developments have been recognised by forecasters, with corresponding implications for the perceived scope of monetary policy.

These trends are largely consistent with alternative survey based measures of the long-run interest rate. For instance, Bauer et al. (2024) highlight a similar decline using Blue Chip Financial Forecast within the context of the Taylor rule and market perceived fluctuations in the policy rate. Benigno et al. (2024) also identify declines in the perceived natural rate, defined as the difference between the long-run expected policy rate and expected level of inflation, using median responses in the Federal Reserve Bank of New York's Survey of Primary Dealers (SPD). These findings also suggest that markets have persistently revised down their expectations of the natural rate, which seems to corroborate established trends in r* during this period.

Finally, we note that the implied natural rate of interest is less sensitive to transitory pressures than the natural rate estimated from realised data. The incremental character of these revisions is consistent with the nature of expectations formation in which assessments of slow-moving fundamentals are revised rather than in response to individual observations. The implied natural rate is accordingly smoother than its benchmark counterpart, a property inherited from inputs which are themselves smoother than the realised series on which the latter is estimated.

[8] Two-sided (smoothed) estimates of the natural rate of interest are presented in Appendix C.

5.3 Heterogeneous Beliefs

Estimates reported thus far are constructed from the mean forecast in each quarter, and therefore describe the perceived natural rate of a representative forecaster rather than the wider distribution from which that average is drawn. If the consensus estimate were an artefact of aggregation, with respondents holding views that bear little relation to the mean, then the misalignments documented above would be a property of the aggregation rather than of market perceptions. The width of the distribution is also informative in its own right, since disagreement about the equilibrium rate is the object that our dispersion weighting exploits, and is therefore useful to observe directly rather than solely through its effect on the filter.

We therefore re-estimate the perceived natural rate separately for each individual respondent to the Survey of Professional Forecasters, retaining those who report a sufficient number of consecutive quarters. We adopt two approaches. In the first, we hold the model parameters at their baseline values and filter each respondent's forecasts through the same system, so that the only source of variation across respondents is the content of their forecasts. In the second, we re-estimate the full three-stage procedure separately for each respondent, so that individuals differ in the parameters governing how forecasts map into an equilibrium rate as well as in the forecasts themselves. Comparing the two isolates the contribution of each channel. The first measures disagreement about the state of the economy, holding the model fixed, while the difference between the two measures the additional dispersion that arises when each respondent is permitted their own model.

Our findings are presented in Figures 4 and 5. Under fixed parameters individual paths cluster closely around the consensus. The majority lie close to the aggregate for most of the sample, and the distribution traces the same secular decline, the same local maxima, and the same post-crisis fall. Dispersion is greatest at either end of the sample, reflecting both genuine disagreement over the disinflation and the smaller number of respondents reporting in the early years. That the spread is modest for most of the period indicates that the secular decline in the perceived natural rate is a property of the distribution of forecasters rather than of its centre alone.

Re-estimating the parameters of the state-space model widens the distribution considerably in the tails. The central mass is comparable to the fixed-parameter case, but a number of individual paths fall below zero in the years after the financial crisis. Since the two exercises share the same forecast data and differ only in whether parameters are re-estimated, this additional dispersion

is largely attributable to parameter heterogeneity. For those respondents whose data imply slope coefficients close to their bound, the implied innovation variance of becomes large and the estimated natural rate can drift far from the consensus. We therefore treat the fixed-parameter estimates as more informative in measuring disagreement about the state of the economy, since they hold the model constant across respondents. The re-estimated version indicates the wider dispersion that arises once respondents are also permitted to differ in the parameters governing their estimates.

5.4 Estimates During the Pandemic

The pandemic presents a difficulty for the estimation of the natural rate of interest; one tempting approach is to omit the period altogether. We therefore compare our dispersion-weighted estimates against two specifications in which the affected observations are excluded from the estimation, the first removing the recession quarters dated by the National Bureau of Economic Research and the second removing the whole of the 2020 calendar year.

The exclusion windows are implemented by setting κ_t to an arbitrarily large value over the affected quarters, which drives the Kalman gain to zero and down-weights those observations from the filter, the smoother, the likelihood and the second-stage regression simultaneously, while leaving the sample timeline contiguous. Our findings are presented in Figure 6, with each exclusion path broken over its respective window.

The choice of window matters considerably. Excluding the full calendar year produces almost no discontinuity, with the series resuming within a few basis points of where it terminates. Excluding the recession quarters alone produces a break of more than a full percentage point, since the shorter window returns the filter to the data in the quarter of reopening and does so with two quarters of accumulated surprise to absorb at once. This is the difficulty that motivates the wider exclusion windows adopted by Holston et al. (2023).

Our own estimates decline by around twenty basis points over the quarter of reopening before recovering over the following year. Since κ_t exceeds two over these quarters, the observations are down-weighted rather than excluded, and the estimate registers the pandemic without being displaced by it. The dispersion adjustment therefore achieves the same continuity as the wider exclusion window while retaining the observations themselves, and does so on the basis of the disagreement observed in the survey rather than an exclusion window chosen in advance.

5.5 Longer-Run Rate Expectations

As an additional check on our estimates, we compare the benchmark and market-perceived natural rate against two direct measures of longer-run policy rate expectations: the median longer-run federal funds rate projection from the Federal Open Market Committee's Summary of Economic Projections (SEP), and the equivalent median longer-run federal funds rate expectation reported in the Federal Reserve Bank of New York's Survey of Primary Dealers (SPD), subsequently supplemented by responses from the Survey of Market Expectations (SME).[9] Unlike our own implied estimates, which are recovered indirectly from the joint behaviour of one-quarter-ahead forecasts of output, inflation and the short-term interest rate, both benchmarks are direct point elicitations of where each panel expects the nominal policy rate to settle over the long run.

Since both series are reported in nominal terms, we convert them to real terms for comparability, subtracting the median longer-run projection for personal consumption expenditures (PCE) inflation reported alongside the SEP's longer-run federal funds rate. We adopt this particular deflator for two reasons. First, it is the same institutional assessment that FOMC participants apply in forming their own longer-run federal funds rate projection, and is therefore internally consistent with the SEP series itself. Second, it has remained effectively static at 2 percent throughout the window we consider, so that it acts as a fixed point of reference rather than an independent forecast that varies meaningfully across survey rounds.[10]

Figure 7 plots the resulting real rate expectations for each survey alongside our natural rate estimates, restricted to the period from January 2012, when the SEP first began publishing quantified longer-run projections, to the end of 2025. Each survey benchmark is plotted only over the months in which a release exists, without interpolation. Both survey measures move closely together, and generally align with our estimates between 2012 and 2016-17, over which period the natural rate of interest and, in particular, perceptions of it, remain comparatively stable and track one another closely throughout.

[9] The Survey of Primary Dealers, later merged into the Survey of Market Expectations, asks respondents to 'provide your estimate of the longer run target federal funds rate.' We use the median response, taken from the Survey of Primary Dealers up to the merger and the Survey of Market Expectations thereafter.

[10] The Survey of Primary Dealers collects its own measure of longer-run inflation expectations intermittently and in a format that has changed across survey vintages. We therefore apply the Summary of Economic Projection's median longer-run PCE projection uniformly to both series, so that the resulting real rates are deflated by a common and time-invariant benchmark rather than by inflation expectations that may differ in basis, horizon or availability across panels. Since this projection has remained at, or within a few basis points of, 2 percent for the duration of the period considered here, the choice of deflator makes little practical difference to the level of either series.

The pandemic produces a marked divergence, with the dealer survey falling towards zero at its trough, before all four series converge again towards the end of the sample. This pattern points to a distinction between stated and implied perceptions of the longer-run equilibrium rate. Whereas the SEP and dealer survey measures are direct, low-frequency elicitations of where each panel expects the policy rate to settle over the long run, our estimates are recovered indirectly, from the extent to which quarter-ahead survey forecasts of output and inflation behave in a manner consistent with a given real rate gap. That our estimates continue to track r* closely through the pandemic, over precisely the period in which stated longer-run expectations diverge sharply, suggests that forecasters became considerably more pessimistic in what they said about the long-run policy rate than they were, at least implicitly, in how they forecast the near-term evolution of the economy. Put differently, the divergence in stated longer-run expectations over the pandemic appears to reflect a revision in sentiment that is not obviously mirrored in the forecasting behaviour from which our own estimate of the perceived natural rate of interest is recovered.

We interpret this comparison with some caution, since the panels underlying each series are not identical. In particular, the SEP is compiled from the assessments of FOMC participants, the dealer survey from primary dealers and other market participants, and our own estimates from the Survey of Professional Forecasters, such that the comparison is across rather than within panels. Nonetheless, the patterns documented here, such as the close alignment at the start of the sample, a marked and sustained divergence from around the start of the pandemic, and renewed convergence towards its end, offers a useful external cross-check on our own estimates, and suggests that the wedge between stated and implied perceptions of the natural rate of interest over this period may merit closer study using direct survey based responses.

5.6 Evidence from Forward Rates

To further investigate how these multivariate estimates compare to alternative univariate measures of the forecast natural rate, we make use of forward rates associated with yields on longer-term treasury security bonds sourced directly from the Federal Reserve of St. Louis (FRED) database. In particular, we compute the difference between the seven-year-ahead zero-coupon one-year yield and the rate of inflation.[11] To decompose the trend component in these rates, we use a range of

[11] Forward rates are computed using the seven-year-ahead zero-coupon one-year yield, based on the fitted yields on eight- and seven-year bonds from Kim and Wright (2005). These are constructed as: $((1 + a/100)^8/(1 + b/100)^7) - 1$. We then average on a quarterly basis and subtract the inflation rate to derive a series for the real forward rate.

conventional statistical techniques, namely the band-pass (BP) filter, the Hodrick-Prescott (HP) filter, and the Stock and Watson (2007) unobserved components with stochastic volatility (UCSV) model. Our findings are presented simultaneously in Figure 8.

Alternative univariate measures using forward rates reveal a more pronounced decline over the sample than our estimates using a multivariate approach, which in contrast exhibits considerably more variation at business-cycle frequencies. This reflects the nature of a semi-structural model that conceives of r-star as a medium-run concept and that separates persistent from transitory movements in the observables. Univariate techniques instead extract the low-frequency component of the real forward rate and are unable to distinguish transitory pressures from movements in the trend. Such techniques perform adequately when inflation and output are relatively stable, but are less reliable over periods in which they are not.

Furthermore, term premia contained within forward rates are often a source of variation that undermine their reliability as a measure of expectations of future rates (Kim and Wright, 2005). Shifts in premia over the period in which the two measures diverge therefore make it difficult to attribute the decline in the filtered forward rate to a revision in perceptions of the equilibrium rate rather than to a change in compensation for duration risk. Survey forecasts are not subject to such contamination, providing further motivation for our approach. Notwithstanding these limitations, univariate measures still yield some indication of market perceptions of the natural rate of interest, whose estimated long-run decline largely corroborates our general findings.

5.7 Market Misalignments

A natural extension of this analysis is to determine the extent to which market participants misperceive the equilibrium interest rate. This information is of use to policymakers, since expectations about the evolution of the natural rate have consequences for the transmission of monetary policy. To achieve this, we simply date the benchmark rate to the quarter to which the forecast refers and compute the difference between the two series. We plot this series in Figure 9.

Our findings document a number of episodes in which market participants either over-predict or under-predict the equilibrium interest rate. After a brief period of overestimation in the mid-1980s, forecasters underestimate the natural rate for two decades. This gap increases through the late 1990s, reaching roughly 160 basis points around the turn of the century, and then narrows over the following decade, closing shortly before the financial crisis of 2008. Following the crisis,

forecasters overestimate the natural rate by as much as 130 basis points, before the two measures closely align from 2015. The pandemic generates a brief and marginal period of underestimation, followed by a return to alignment towards the end of our sample.

Given our knowledge of professional forecasts of the real interest rate, presented in Figure B.4, periods in which participants overestimate the natural rate of interest after 2008 are predominantly characterised by declining real rate forecasts. Since the benchmark neutral rate falls sharply during the Great Recession, markets therefore judge the prevailing stance to be more accommodative than the data imply. The reverse holds in the mid-to-late 1990s, when markets underestimate the natural rate while expecting positive real rates. As the benchmark rate rises substantially over this period, participants judge policy to be more restrictive than it was.

Widening gaps between these two series are driven primarily by shifts in the natural rate of interest. For instance, the period of underestimation in the late 1990s coincides with the productivity boom, over which realised output growth implied a rising equilibrium rate that market perceptions seem to track poorly. In addition, overestimation in the post-crisis period follows a precipitous and well documented reduction in the natural rate, to which forecasts seem to adjust far more gradually. These findings suggest that whilst market expectations generally move with the underlying natural rate of interest, the response is largely attenuated and confined to its secular path.

5.8 Fundamental Drivers

Market beliefs of the natural rate of interest are likely driven by a wide range of factors. Beyond the equilibrium interest rate, it is reasonable to suggest that market participants base their estimation of the future evolution of r-star on changes in macro fundamentals. In this regard, we investigate how these beliefs respond to shifts in consumer price inflation, real gross domestic product, and short-term nominal interest rates. We also consider the contribution of the natural rate of interest itself, since changes in the underlying object professional forecasters are implicitly assessing are likely to inform their view of where the equilibrium lies.

To proceed with our analysis, we use local-projection (LP) methods and a vector autoregression (VAR) model to estimate the response of the implied natural rate to shocks in these variables. We adopt this dual approach to verify the robustness of our results due to recent discussions concerning both the similarities and differences in population and finite samples between these models (e.g. Plagborg-Møller and Wolf, 2021; Li et al. 2024), particularly in the estimation of impulse

response functions. Our approach is also informed by the mixed empirical evidence of stationarity in our series, as presented in Table G.1. Johansen tests nonetheless identify at least one long-run relationship between the variables in this system, which motivates our approach to estimate both models in levels rather than imposing a specific cointegrating structure.

For the local projections (LP), we follow the standard approach of Jordà (2005), with optimally chosen lag lengths using the Akaike Information Criteria (AIC). For the vector autoregression (VAR), we identify shocks using a Cholesky decomposition with the order: interest rate, inflation, output, natural rate, and expected natural rate. The justification for this order is somewhat intuitive and assumes that forecasts of the natural rate of interest are driven by variables in descending exogeneity. Given this system contains two estimated series, we interpret the direction and persistence of the responses rather than their specific point magnitudes.

Estimates of the impulse response function (IRF) for each model are presented in Figures 10-11. Our results from Local-Projections show that unit shocks to output create a positive response in perceptions of the natural rate. We observe a slow decay in the response to output suggesting a high level of persistence. Shocks to consumer price inflation initially generate a positive response in the implied natural rate, although this response declines over longer horizons. As regards shocks in estimates of the equilibrium rate itself, we generally observe a positive response in forecasts of the natural rate of interest, which seems to exhibit some persistence over more longer horizons. This implies that market participants revise their expectations of r* in the same direction as changes in the underlying measure they are forecasting.

In all but magnitude, estimates of the impulse response function from our VAR framework are broadly similar to those estimated using local-projections. To investigate the individual contributions of these fundamentals across time, we compute the historical decomposition of the forecast natural rate arising from the model, which is presented in Figure 12. Our results suggest that inflation, interest rates, and output do play a role in explaining the historical variation in perceptions of the natural rate of interest. The nominal interest rate initially makes the smallest contribution to perceptions of the natural rate. Its contribution increases substantially around the turn of the century and remains sizeable until the financial crisis, before declining thereafter. Output makes a relatively small contribution throughout the sample, although its contribution increases around the pandemic period. Inflation contributes substantially during the 2000s, but its contribution declines sharply following the financial crisis. Notably, the underlying natural rate itself makes its strongest

contribution in the period immediately following the Great Recession, coinciding with the period in which the natural-rate concept gained considerable prominence in monetary-policy discussions.

5.9 Impact in Financial Markets

Existing research establishes a link between the natural rate of interest and wider financial conditions (Kiley, 2015; Cukierman, 2016, Taylor and Wieland, 2016). Recent literature has focused on incorporating important financial pressures into estimates of the equilibrium real interest rate. For instance, Akinci et al. (2023) propose the concept of the financial stability interest rate, which reflects the real interest rate consistent with financial, rather than macro stability. Using a macro-financial model with occasionally binding credit constraints, the authors show how real rates affect financial conditions asymmetrically over time. In the short run, interest rate hikes can trigger balance sheet stress via valuation effects, whilst in the medium run, persistently low rates encourage risk-taking and asset reallocation, increasing system vulnerability. These findings imply that monetary policy cannot rely solely on r-star, as the financial stability real rate can embed the market's capacity to absorb shocks. This introduces an important trade-off wherein the real rate that stabilises inflation and output may come at the cost of greater financial fragility.

Reis (2022) argues that conventional measures of r-star relying on long-term government bond yields offers an incomplete view of macro-financial conditions, overlooking the stable or rising return on private capital. Using a cross-country approach, the author shows that whilst government bond yields have declined over the past two decades (suggesting a falling natural rate of interest), returns on private investment have remained constant if not marginally increasing. This divergence implies that financial markets, particularly private investment, operate under an alternative natural rate - one that is not captured by public bond yields. Reis (2022) identifies both financial frictions and capital misallocation as key factors driving this wedge and warns that relying on bond-based r-star may lead to misleading policy conclusions, particularly regarding investment behaviour.

In related work, Caballero et al. (2025) propose a financial conditions natural analogue to the natural rate of interest, arguing that the stance of monetary policy is more accurately reflected by a broader set of financial variables, than by the policy rate alone. The authors develop a measure of the natural level of financial conditions that would be consistent with output at potential and inflation at target. This rate is shown to be significantly more stable than traditional estimates of the natural rate, particularly during periods of financial stress such as the global financial crisis.

Whilst the natural rate declines sharply in response to asset price movements, the natural level of financial conditions remains anchored to macroeconomic fundamentals, indicating that it may offer a more reliable signal for monetary policy. These estimates also closely track optimal policy benchmarks derived from counterfactual simulations, except during major crises.

Given that financial pressures have implications for the natural rate, it is reasonable to assume that variation in expectations of r-star also have meaningful consequences across financial markets. In particular, variation in expectations of the natural rate may influence how market participants interpret central bank decisions, shape yield curves, and affect equity valuations. In this regard, r-star is not merely an outcome of macro-financial forces but an input into the financial sector itself. Understanding this two-way interaction is therefore important to assessing the role of expectations in the transmission of monetary policy. In light of this, we investigate this relationship using local projection methods between our survey-based estimates of the expected equilibrium interest rate and measures of key financial metrics across the bond and equity market.

We source quarterly data on the ten-year two-year treasury term spread from the Federal Reserve Bank of St. Louis (FRED), the price-to-earnings ratio from Shiller (2000), the credit-to-GDP gap from the Bank for International Settlements (BIS), and the Adjusted National Financial Conditions Index (ANFCI) from the Federal Reserve Bank of Chicago.[12] In order to appreciate the general impact of variation in the forecast natural rate of interest, these variables are chosen for their wide coverage of the financial sector, and to also complement the medium- to long-term frequency of our semi-structural estimates. For instance, the price-to-earnings ratio we implement is the cyclically adjusted ratio of stock prices to inflation-adjusted (moving) average earnings, which captures future equity returns over longer horizons than traditional price-earnings measures. The Adjusted National Financial Conditions Index complements these by summarising conditions across money, debt and equity markets in a single measure, purged of variation attributable to the current state of the real economy.[13] We also choose to focus on the gap between the market-perceived natural rate of interest and r-star, in order to assess the response of financial markets to forecast errors, decomposing it into its positive and negative components so that the response to underestimation is estimated separately from the response to overestimation.

[12] The price-to-earnings ratio and credit-to-GDP gap both play central roles in a wide range of modern macro-finance models within the existing literature, see, inter alia, Cochrane (2016), Jordà et al. (2017), and Gabaix et al. (2025).

[13] The Adjusted National Financial Conditions Index is published weekly, which we aggregate to quarterly means. The adjusted series removes variation in financial conditions attributable to contemporaneous output and inflation.

We find no evidence from Johansen tests of cointegration in this system (refer to Table G.2 for unit root tests). To proceed we use local projection methods to estimate the impulse responses, and present our findings in Figure 13. Innovations here should be understood as misalignment shocks, in which market expectations of r-star fall short of (exceed) the natural rate of interest itself. We define these as periods of implicit underestimation (overestimation) of the future equilibrating real interest rate. We decompose the misalignment into its positive and negative components

$$\tilde{r}^{*_e}_{t+1} = \max\left(\pm\left(E_t r^*_{t+1} - r^*_{t+1}\right),\, 0\right) \tag{20}$$

since a shock to the gap as a whole carries no direction; a unit innovation is otherwise the same object whether it widens an existing overestimation or underestimation, so that the estimated response averages across two episodes which differ in their content. Estimating these components separately allows each response to be attributed to a direction of misperception.

The four responses are mutually consistent, and describe a loosening of financial conditions following a period of underestimation.[14] The ANFCI is negative upon impact and recovers gradually over longer horizons. Equity valuations rise upon impact, with the cyclically adjusted price-to-earnings ratio significantly positive at each horizon. The credit-to-GDP gap also expands steadily over the medium-run, and the term spread narrows, though this response is significant in the short-run. Taken together, these estimates describe a market that, believing the natural rate to be lower than it is, reads the prevailing stance of policy as more restrictive than it is, anticipates easing, and accordingly discounts future earnings at a lower rate while borrowing more freely.

These findings demonstrate the impact that misalignments between perceptions of the natural rate and r* have on financial markets. Our estimates using professional survey forecast data are therefore useful in determining whether financial market participants have been overly optimistic or pessimistic in their judgments about the future of the natural rate of interest, and the influence these assessments might have on the bond and equity markets within the financial sector. Monetary policymakers must, therefore, also account for market expectations of the equilibrium rate in order to avoid compromising financial stability and improve the transmission of monetary policy when responding to price and output instability in the real sector.

[14] Responses to overestimation are largely the mirror image of those reported here, carrying opposite signs throughout. The two are not of equal magnitude, however, with equity valuations and the credit gap responding more strongly to periods of underestimation and the term spread responding more strongly to periods of overestimation.

6 Conclusion

This paper examines how professional forecasters perceive the natural rate of interest, the benchmark against which the stance of monetary policy is judged. Applying the semi-structural model underlying the Federal Reserve Bank of New York's benchmark to more than four decades of survey forecasts, with observations weighted by the degree of forecaster disagreement, we recover a market-perceived natural rate that can be compared directly with the benchmark estimated from realised data. Our findings are summarised as follows.

First, we detect substantial time variation in market-based perceptions of the natural rate of interest. Our results highlight a secular decline in natural rate forecasts across the historical sample, amounting to some 250 basis points from its mid-1980s peak. Weighting by dispersion allows the pandemic to be retained without the discontinuities that arise when it is omitted, whose magnitude depends on where the exclusion window is drawn. Longer-run policy rate expectations that market participants report directly in central bank surveys are initially aligned with our estimates but diverge sharply around the pandemic, which suggests that respondents lowered their stated view of the long-run policy rate by more than their own forecasts of the economy implied.

These empirical findings are broadly robust across a series of checks. Disaggregating the underlying survey panel, we show that estimates of the perceived natural rate exhibit similar trends at both the consensus and individual level. Comparisons against the median longer-run policy rate expectations reported directly in central bank surveys of market participants, further corroborate the pattern of decline we identify. These forecasts are initially aligned with our own one-sided estimates, but diverge sharply during the pandemic, suggesting that forecasters became considerably more pessimistic in what they reported about the long-run policy rate over this period than they were, at least implicitly, in how they forecast the evolution of the economy.

Second, we document periods in which markets persistently over- or under-predict r*. Absolute deviations from the benchmark natural rate are largest in periods when the natural rate was changing substantially. Forecasters underestimate the natural rate throughout the long expansion preceding the financial crisis, particularly as trend productivity growth strengthened in the late 1990s, and overestimate it during the decline following the crisis. The two measures subsequently converge towards the end of our sample. In each episode, perceptions move in the same direction as the benchmark rate but by a considerably smaller amount, so that market participants capture

the secular trend in r* rather than its short-run volatility. Our dynamic analysis is consistent with this pattern. Innovations in output, inflation and the short-term interest rate all contribute to revisions in the perceived natural rate. The response to innovations in r* itself is the largest of these, but it builds over several years rather than occurring upon impact. This slow adjustment explains why misalignments of the magnitude and duration we document are able to persist.

Third, we show that implied expectations of the natural rate play a significant role in shaping financial market outcomes. We find that underestimations of the natural rate of interest can trigger systematic responses across key financial indicators. In particular, we find that forecast errors lead to a loosening of financial conditions, a sustained rise in equity valuations, and an expansion of the credit-to-GDP gap, with the response of the term spread weaker and confined to the shortest horizons. Periods of overestimation carry the opposite sign throughout. These results suggest that forecast misalignments may influence perceptions of the stance of monetary policy, yield curve dynamics, and risks of financial market instability.

These conclusions have significant implications for the transmission of monetary policy and wider macroeconomic stability. In particular, they imply that the impact of monetary policy upon the market cannot be fully appreciated without also taking into consideration the perceptions that participants have about the future natural rate of interest. More importantly, these beliefs can differ from the benchmark for years, with consequences for asset prices and credit that bear on financial stability. A central bank that communicates the path of its policy rate without communicating its view of equilibrium leaves room for the stance of policy to be misread.

These findings suggest two avenues for further research. Given that market perceptions of the natural rate may diverge from the benchmark for extended periods, the role of central bank communication, including the publication of longer-run projections for the policy rate, in shaping the perceived equilibrium warrants further investigation. The gradual adjustment of these beliefs also raises the question of how monetary policy ought to be conducted when private-sector assessments of equilibrium respond only slowly to changes in the natural rate itself.

Table 1. Survey Forecast Summary Statistics

	Mean	Std. Dev	Skew	Min	Max	Obs
CPI Inflation	2.79	1.43	-0.20	-2.45	7.60	174
3-Month Interest Rate	3.60	2.78	0.29	0.05	10.24	174
Real GDP Growth	2.24	3.60	-7.81	-38.21	17.64	174

Note: Summary statistics for the Survey of Professional Forecasters data entering the state-space system, from 1982:3 to 2025:4. Consumer price inflation and the three-month interest rate are reported as annual rates; output growth is the quarterly log difference of the chained forecast level series, expressed annually.

Table 2. Parameter Estimates

Parameters	Actual	Forecast
λ_g	0.28	0.15
λ_z	0.02	0.02
$\Sigma\alpha_y$	0.92	0.97
α_r	-0.07	-0.08
	(0.02)	(0.02)
β_y	0.06	0.03
	(0.03)	(0.01)
$\sigma_{\tilde{y}}$	0.44	0.46
σ_π	0.79	0.80
σ_{y*}	0.50	0.65
σ_g	0.14	0.10
σ_z	0.13	0.14
σ_{r*}	0.19	0.17
Average Standard Errors		
r^*_{avg}	1.79	2.54
y^*_{avg}	1.31	1.60
g_{avg}	0.52	0.21

Note: Final stage parameter estimates of the state space system for the United States from 1982:3-2025:4. Average standard errors of the natural rate, potential output, trend growth rate, and their forecast analogues are reported in the last three rows. The standard deviation σ_{r*} is calculated as the square root of $\sigma_g^2 + \sigma_z^2$.

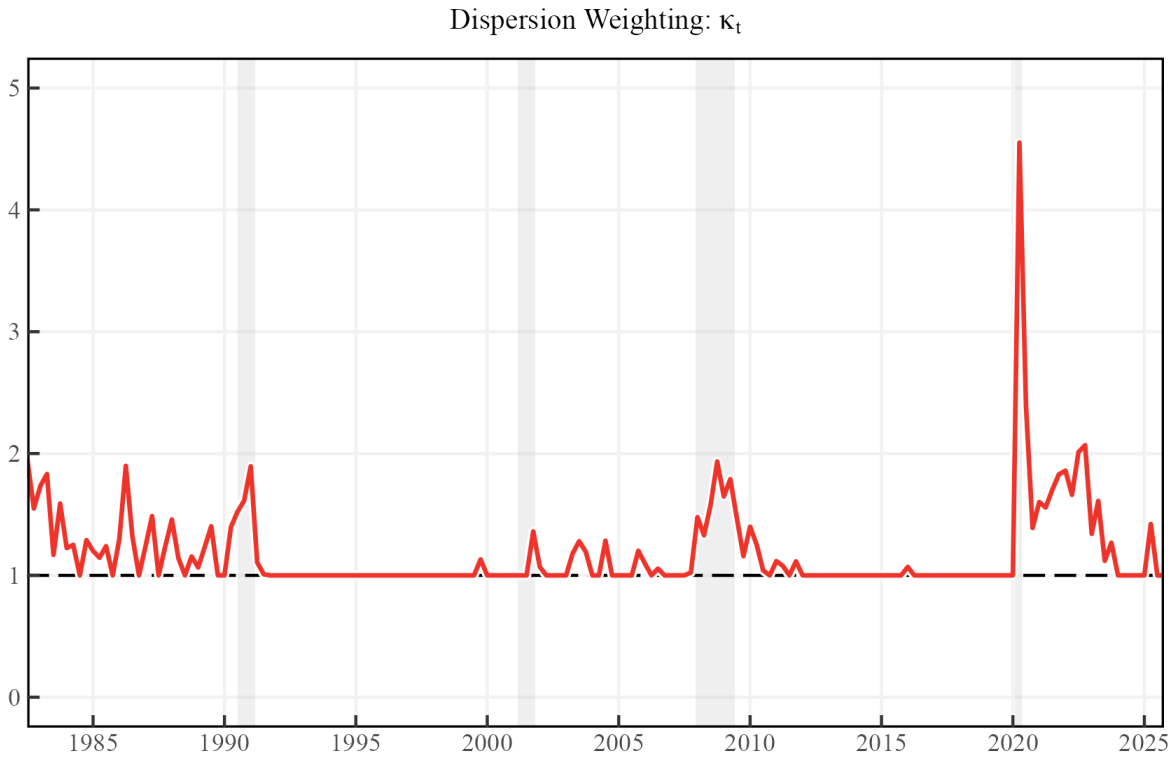


Figure 1. Dispersion Weighting

Note: The dispersion weighting factor κ_t, constructed as the geometric mean of the normalised interquartile ranges of output and inflation forecasts, floored at unity. Values greater denote quarters in which forecaster disagreement exceeds its pre-pandemic average and the corresponding observation is down-weighted in the filter. The dashed line marks $\kappa_t = 1$, at which the model reduces to its unadjusted counterpart. Shaded regions in grey denote recessionary periods sourced directly from the National Bureau of Economic Research.

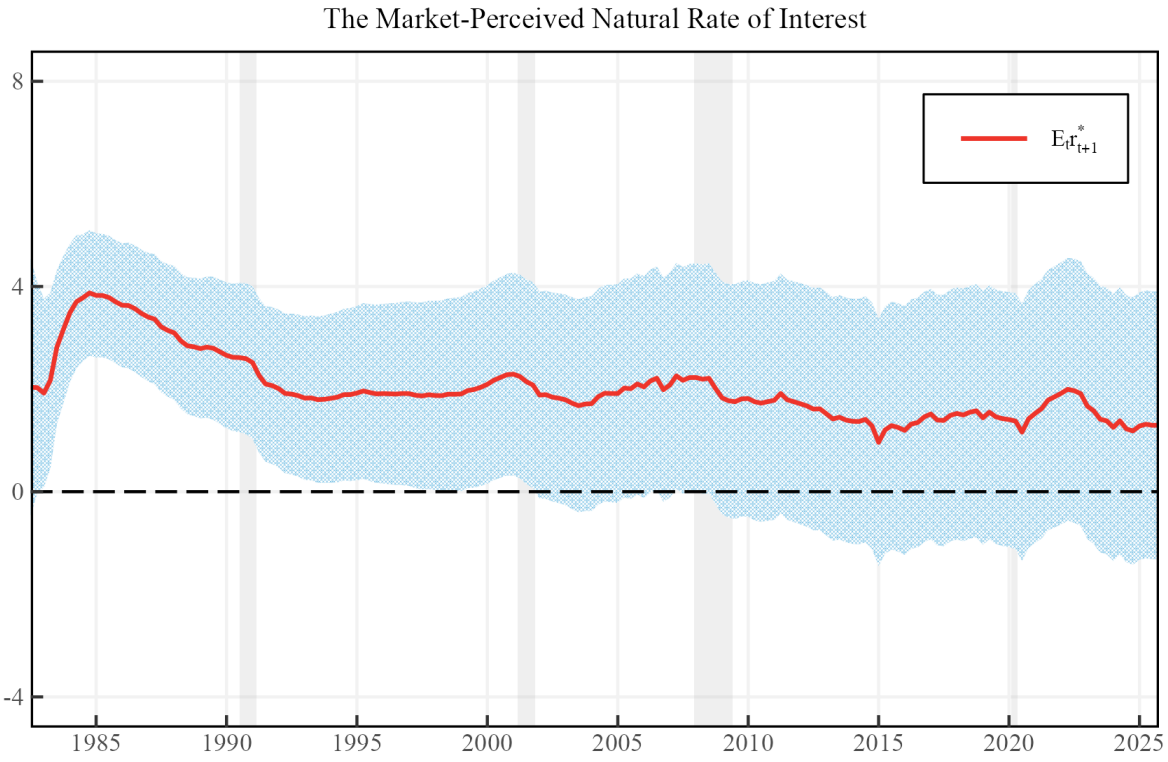


Figure 2. The Market-Perceived Natural Rate of Interest

Note: One-sided Kalman filter estimates of the market-perceived natural rate between 1982:3 and 2025:4. Obtained from applying the dispersion-weighted state-space model to survey forecast data. Shaded regions in blue represent the 90% confidence band.

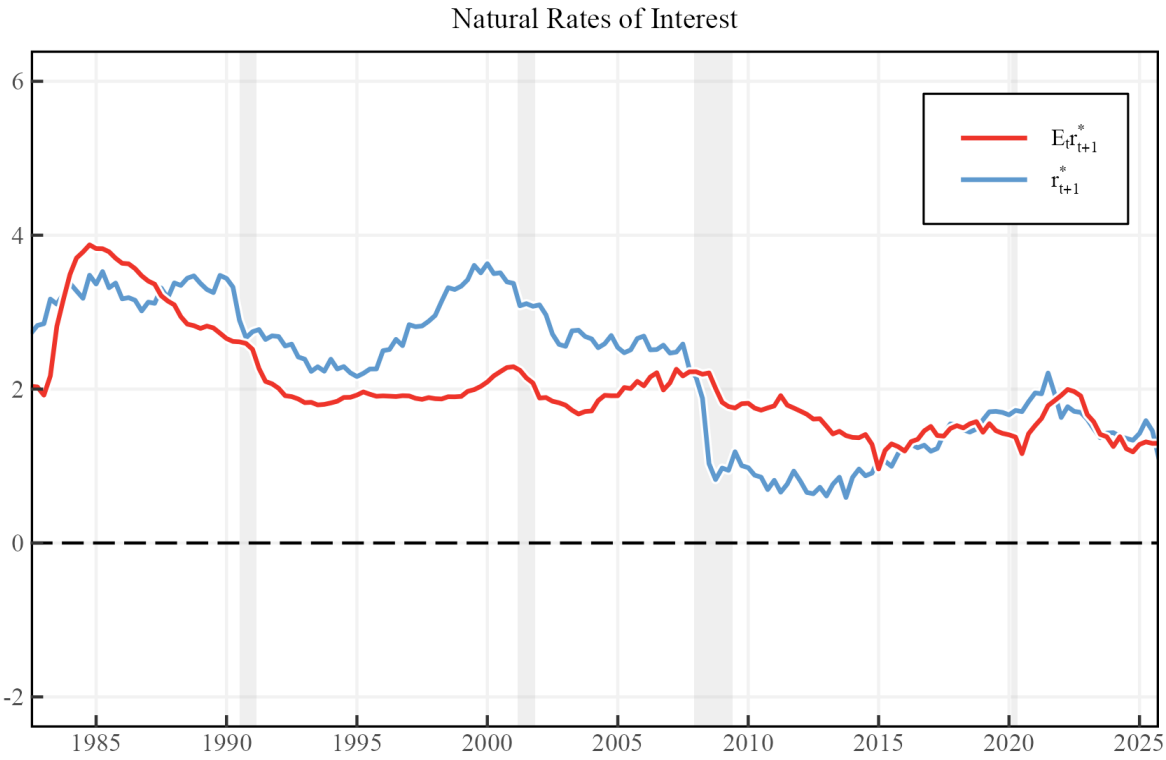


Figure 3. Natural Rates of Interest

Note: One-sided Kalman filter estimates of the market-perceived (red) and benchmark (blue) natural rate of interest from 1982:3 to 2025:4. The former series is obtained by applying the dispersion-weighted state-space model to survey forecast data; the latter series arises from applying the Holston et al. (2023) adjusted model to realised data, shifted forward one quarter to ensure alignment between series.

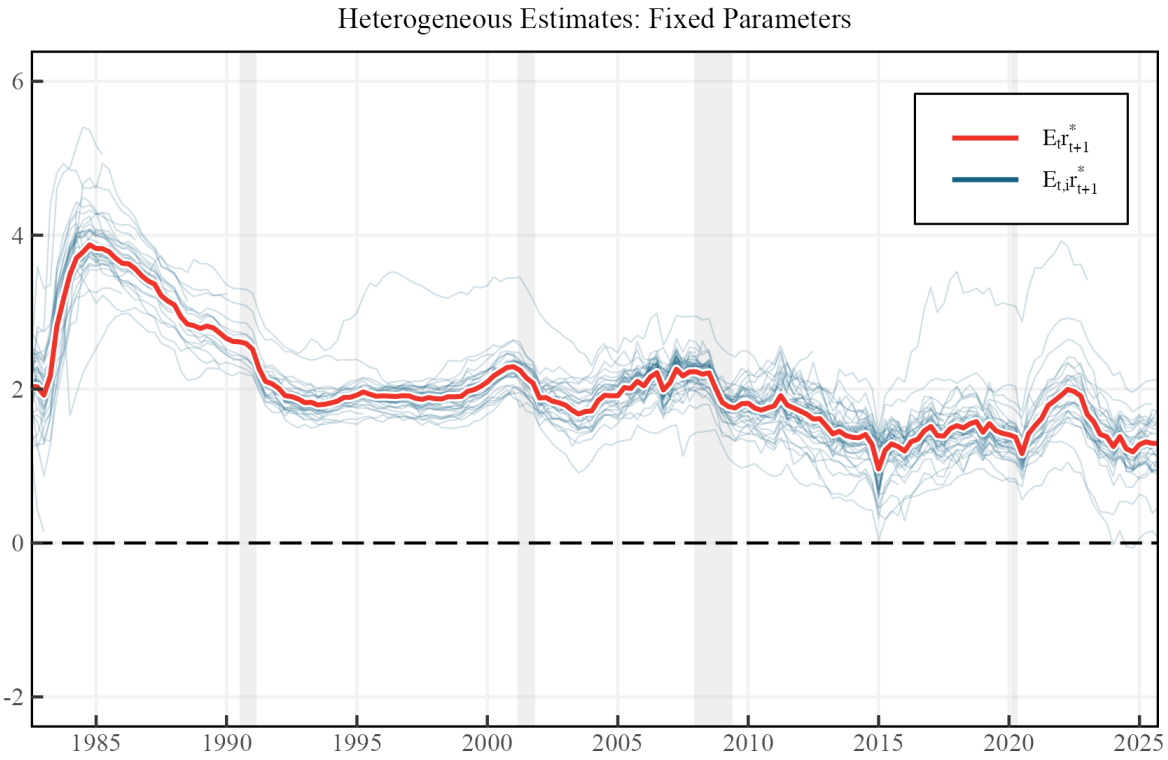


Figure 4. Heterogeneous Estimates: Fixed Parameters

Note: One-sided estimates of the perceived natural rate for each individual respondent to the Survey of Professional Forecasters, shown in blue, against the consensus estimate in red. The model parameters are held at baseline values. Individual paths are plotted only over the quarters in which the respondent reports.

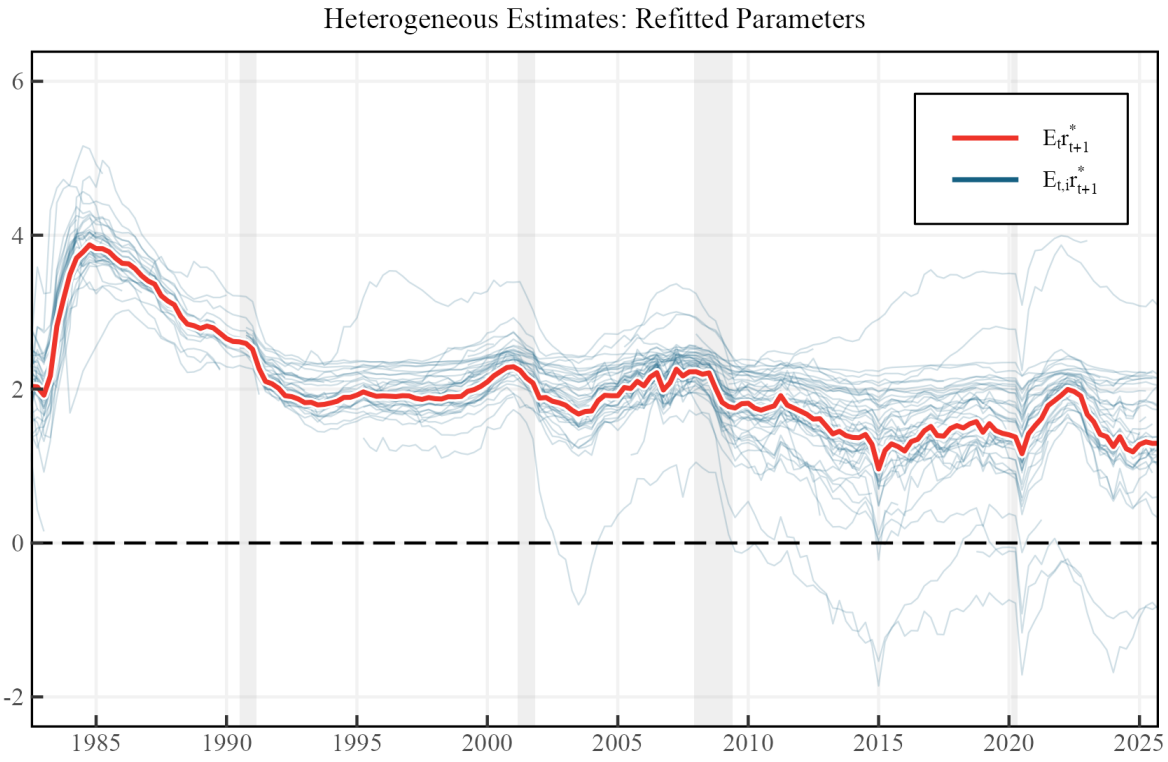


Figure 5. Heterogeneous Estimates: Refitted Parameters

Note: One-sided estimates of the perceived natural rate for each individual respondent to the Survey of Professional Forecasters, shown in blue, against the consensus estimate in red. Parameters are re-estimated for each respondent. Individual paths are plotted only over the quarters in which the respondent reports.

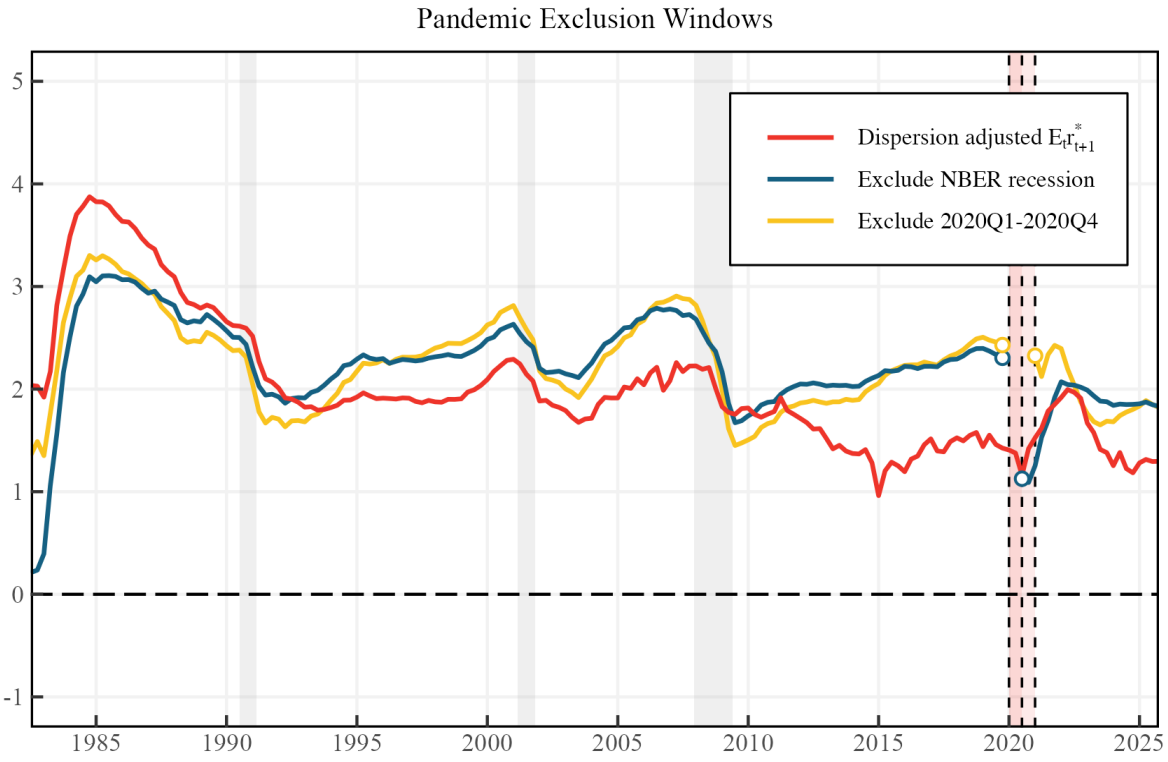


Figure 6. Pandemic Exclusion Windows

Note: One-sided estimates of the market-perceived natural rate under varying exclusion windows. Shaded vertical regions and dashed lines mark the exclusion windows, the narrower corresponding to the recession quarters dated by the NBER (2020:1-2020:2) and the wider to the year 2020. Baseline estimates are in red.

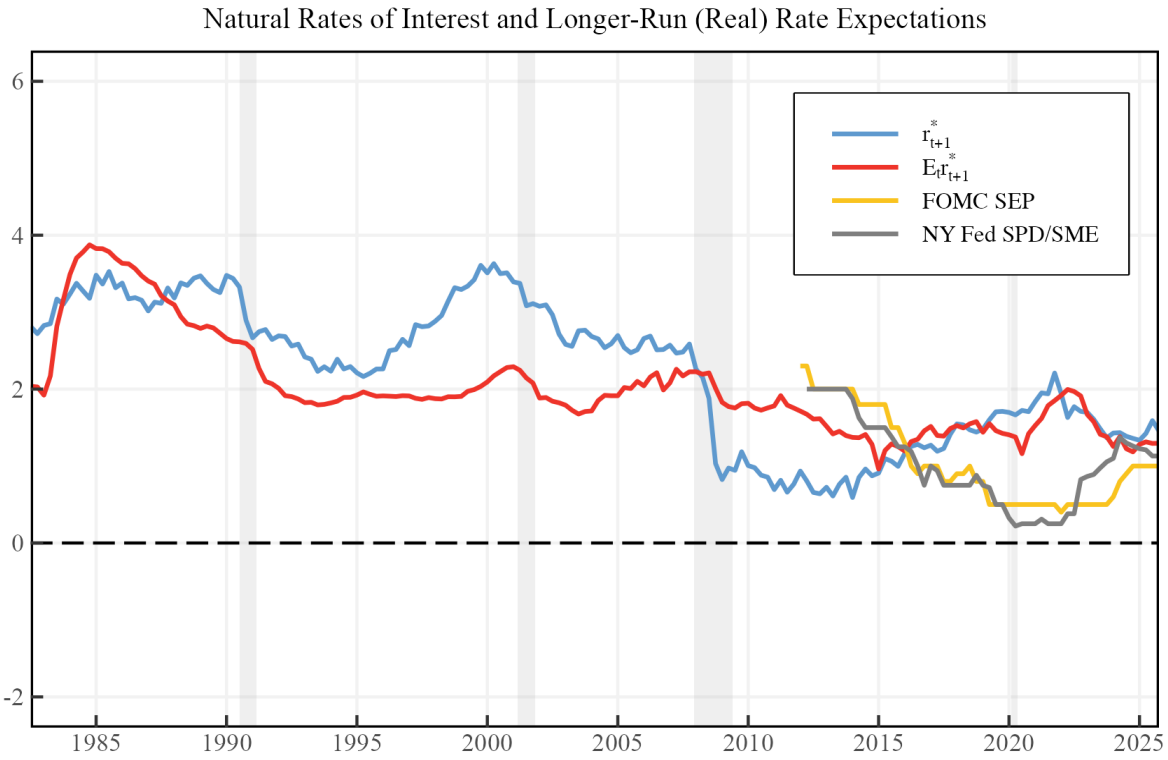


Figure 7. Natural Rates of Interest and Longer-Run (Real) Rate Expectations

Note: One-sided Kalman filter estimates of the market-perceived (red) and benchmark (blue) natural rate of interest, alongside the median longer-run federal funds rate expectation from the Federal Open Market Committee's Summary of Economic Projections (yellow) and the Federal Reserve Bank of New York's Survey of Primary Dealers/Survey of Market Expectations (grey), from January 2012 to December 2025. Both survey benchmarks are deflated by the Summary of Economic Projections' median longer-run projection for personal consumption expenditures inflation, which has remained at approximately 2 percent throughout the sample. See Figure E.1 for a more detailed visual comparison over the period in which all series exist.

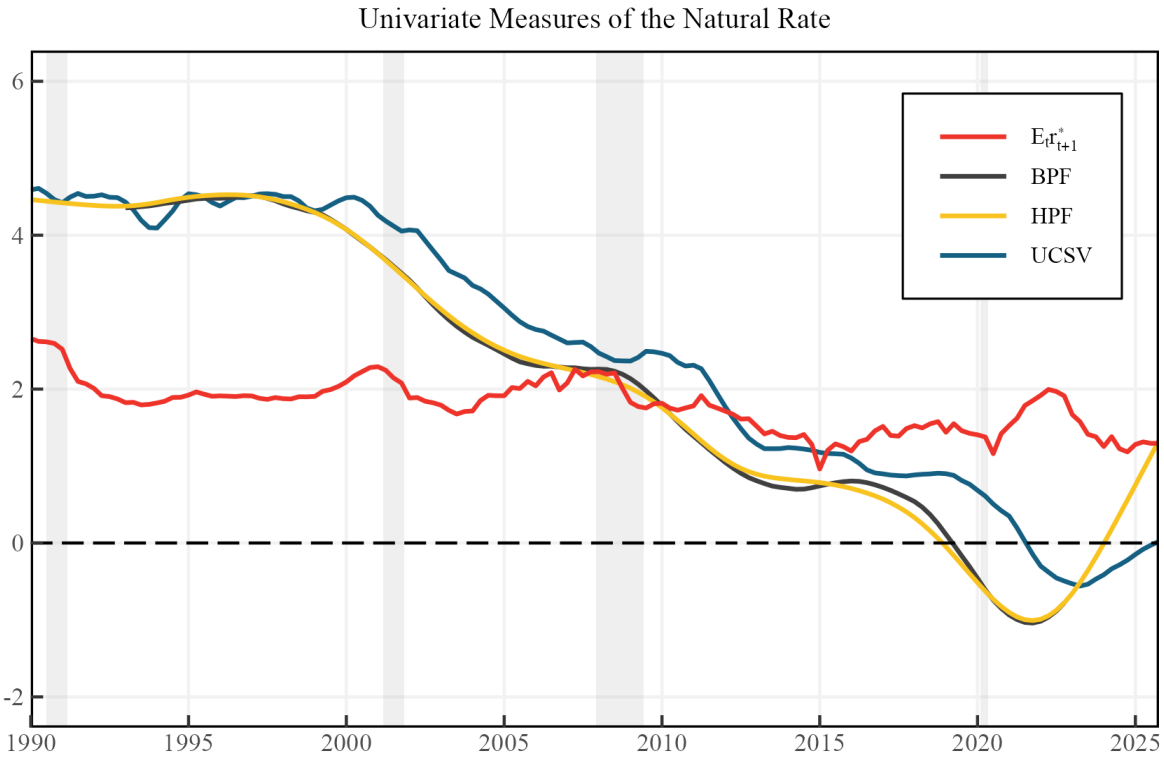


Figure 8. Univariate Measures of the Natural Rate

Note: Alternative univariate measures of the perceived natural rate of interest from 1990:1 to 2025:4. One-sided (multivariate) Kalman filter baseline estimates are presented in red. Band-pass filter (BPF) estimates are presented in black. Hodrick-Prescott filter (HPF) estimates are presented in yellow. Estimates from the unobserved components model with stochastic volatility (UCSV) are presented in blue.

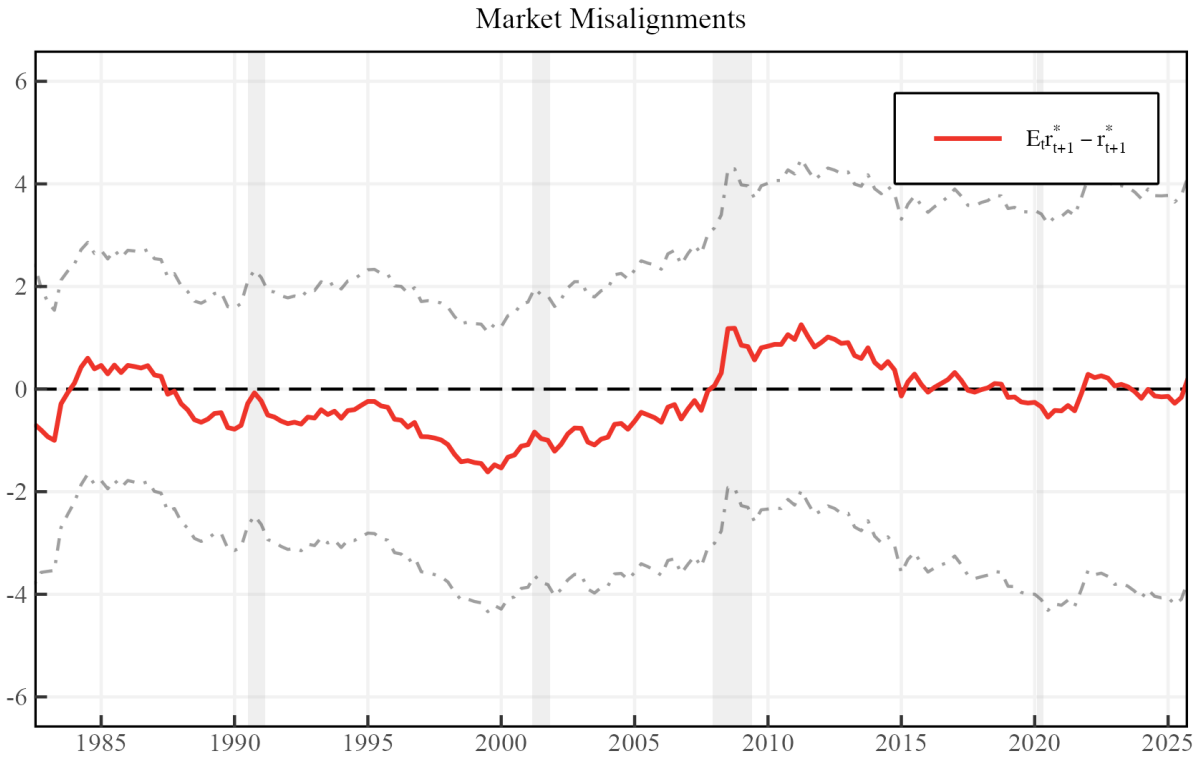


Figure 9. Market Misalignments

Note: Deviations between estimates of the market-perceived natural rate of interest $E_t r^*_{t+1}$ and the natural rate of interest r^*_{t+1} between 1982:3 and 2025:4. Positive values are associated with periods of overestimation, in which $E_t r^*_{t+1} > r^*_{t+1}$, and negative values with underestimation. The dot-dashed line represents the 90% confidence band, constructed by combining the one-sided standard errors of the market-perceived and benchmark natural rates. The band assumes the estimation errors of the two series are independent.

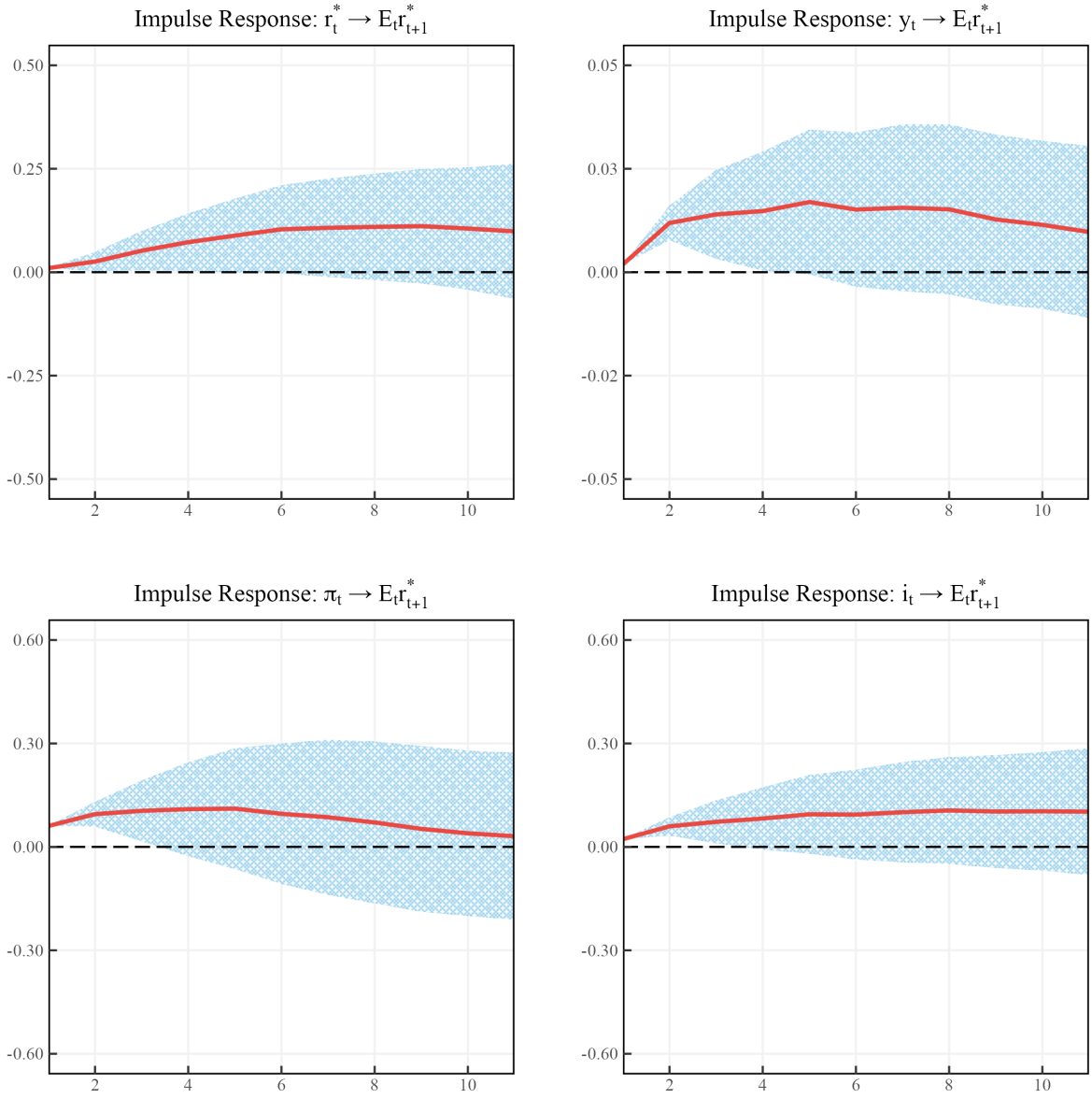


Figure 10. LP Impulse Response Functions

Note: Local projection impulse response functions of a unit innovation in the natural rate of interest r_t^*, real gross domestic product y_t, consumer price inflation π_t, and the short-term (three-month) nominal interest rate i_t, on the perceived natural rate of interest $E_t r_{t+1}^*$. Shaded regions represent the 90% confidence band.

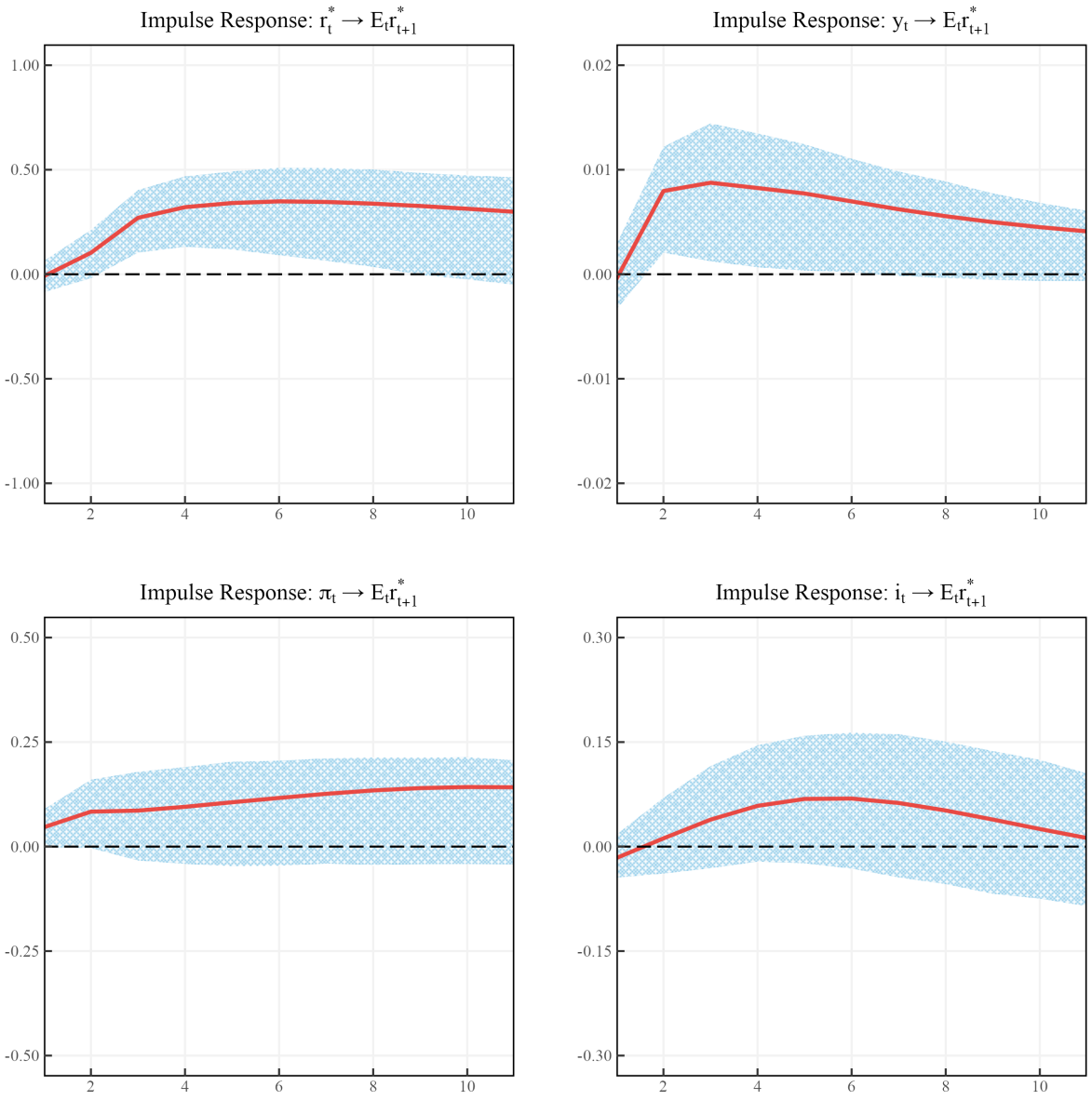


Figure 11. VAR Impulse Response Functions

Note: Vector autoregression impulse responses of an innovation in the natural rate of interest r_t^*, real gross domestic product y_t, consumer price inflation π_t, and the short-term (three-month) nominal interest rate i_t, on the perceived natural rate of interest $E_t r_{t+1}^*$. Shaded regions represent the 90% confidence band.

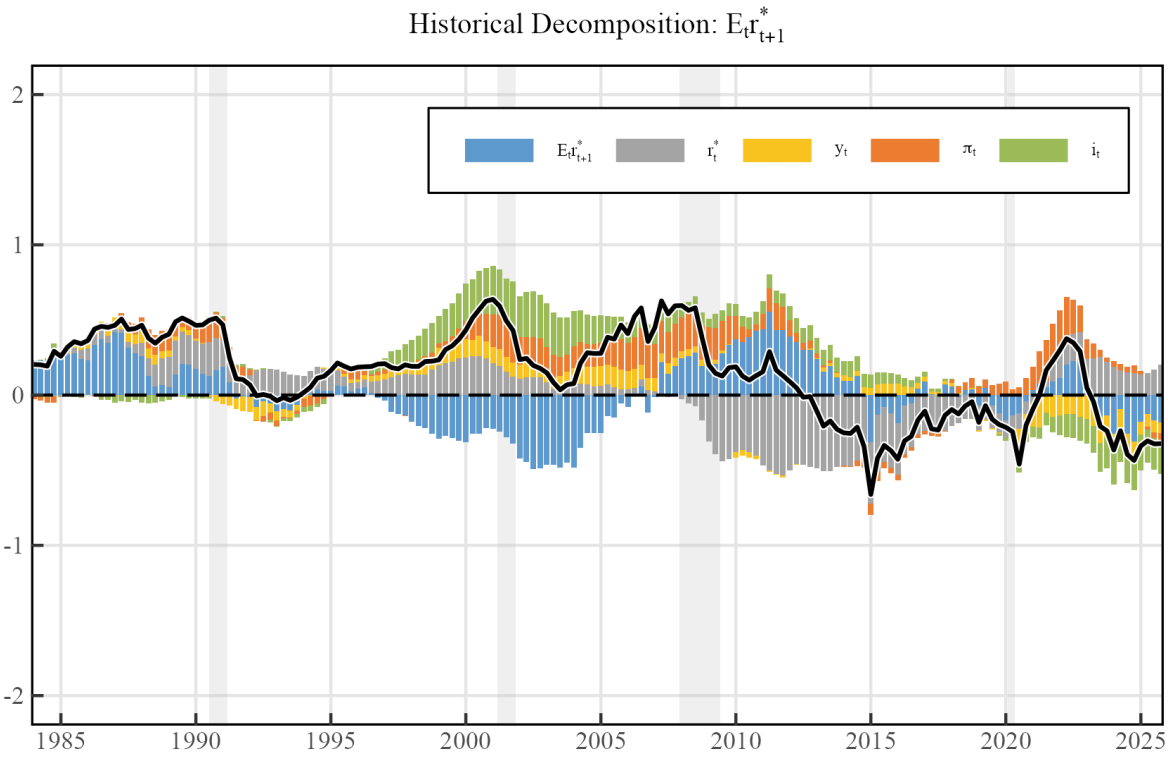


Figure 12. VAR Historical Decomposition

Note: Vector autoregression historical decomposition of the perceived natural rate $E_t r^*_{t+1}$ between 1982:3 and 2025:4. Shaded bars are associated with individual contributions of innovations in each variable to the perceived natural rate. The stochastic component to which these shocks aggregate is represented in black. Refer to Figure F.1 for the forecast error variance decomposition associated with the implied natural rate.

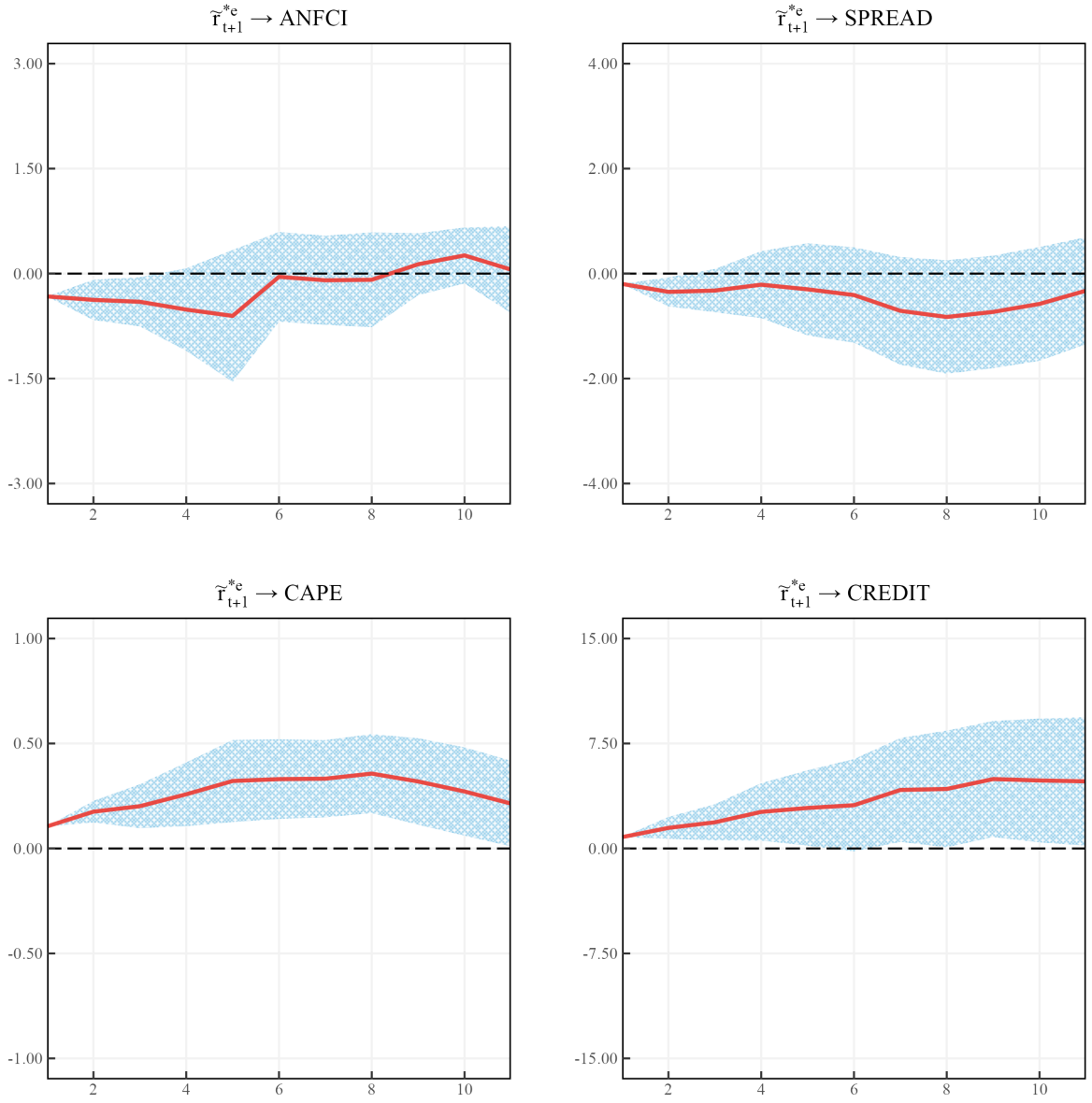


Figure 13. Impulse Response Functions

Note: Local projection impulse response functions of a positive innovation in the underestimation component of the gap between the implied and benchmark natural rate of interest $\tilde{r}^{*e}_{t+1} = \max\left(r^{*}_{t+1} - E_t r^{*}_{t+1}, 0\right)$*, on the Adjusted National Financial Conditions Index (ANFCI), the ten-year two-year treasury term spread (SPREAD), the cyclically adjusted price-to-earnings ratio (CAPE), and the credit-to-gross domestic product gap (CREDIT). The overestimation component is included in the system throughout, and its responses are broadly the mirror image of those reported here. Shaded regions correspond to the 90% confidence band.*

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APPENDIX

The Market-Perceived Natural Rate of Interest

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King's College London	University of Sheffield	Wilfrid Laurier University

A Model Specification

We estimate the system in the three stages of Holston et al. (2017), in the state-space notation of Hamilton (1994). The observation and state equations are

$$\mathbf{y}_t = \mathbf{A}'\mathbf{x}_t + \mathbf{H}'\boldsymbol{\xi}_t + \mathbf{w}_t, \qquad \boldsymbol{\xi}_t = \mathbf{F}\,\boldsymbol{\xi}_{t-1} + \mathbf{v}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \;\; \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{21}$$

where the observables are one-quarter-ahead survey forecasts and $\pi_{t-1,3|t-2,4}$ denotes the average of the second through fourth lags of forecast inflation. The dispersion-weighted adjustment makes the observation covariance time-varying,

$$\mathbf{R}_t = \kappa_t^2\,\mathbf{R}, \qquad \mathbf{R} = \operatorname{diag}\!\left(\sigma_{\tilde{y}}^2,\, \sigma_{\pi}^2\right), \qquad \kappa_t \geqslant 1, \tag{22}$$

with κ_t constructed from forecaster dispersion as in Section 4. Setting $\kappa_t = 1$ recovers the unadjusted specification. In every stage the log-likelihood is the prediction-error decomposition (19) with the κ_t-scaled innovation covariance $\mathbf{S}_t = \mathbf{H}'\mathbf{P}_{t|t-1}\mathbf{H} + \kappa_t^2\,\mathbf{R}$.

Stage I Specification:

$$\mathbf{y}_t = \begin{bmatrix} y_{t+1|t}, & \pi_{t+1|t} \end{bmatrix}' \qquad \mathbf{x}_t = \begin{bmatrix} y_{t|t-1}, & y_{t-1|t-2}, & \pi_{t|t-1}, & \pi_{t-1,3|t-2,4} \end{bmatrix}'$$

$$\boldsymbol{\xi}_t = \begin{bmatrix} y^*_{t+1|t}, & y^*_{t|t-1}, & y^*_{t-1|t-2} \end{bmatrix}'$$

$$\mathbf{H}' = \begin{bmatrix} 1 & -\hat{\alpha}_{y,1} & -\hat{\alpha}_{y,2} \\ 0 & -\hat{\beta}_y & 0 \end{bmatrix} \qquad \mathbf{A}' = \begin{bmatrix} \hat{\alpha}_{y,1} & \hat{\alpha}_{y,2} & 0 & 0 \\ \hat{\beta}_y & 0 & \hat{\beta}_\pi & 1-\hat{\beta}_\pi \end{bmatrix}$$

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad \mathbf{Q} = \begin{bmatrix} \sigma_{y*}^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \mathbf{R}_t = \kappa_t^2 \begin{bmatrix} \sigma_{\tilde{y}}^2 & 0 \\ 0 & \sigma_{\pi}^2 \end{bmatrix}$$

Vector estimated by maximum likelihood:

$$\theta_1 = \begin{bmatrix} \hat{\alpha}_{y,1}, & \hat{\alpha}_{y,2}, & \hat{\beta}_\pi, & \hat{\beta}_y, & g, & \sigma_{\tilde{y}}, & \sigma_\pi, & \sigma_{y*} \end{bmatrix}$$

The signal-to-noise ratio $\lambda_g = \sigma_g/\sigma_{y*}$ is obtained from the Stock and Watson (1998) median-unbiased estimator applied to the structural-break statistic of the smoothed forecast potential-output path. The break regressions at this stage are unweighted.

Stage II Specification:

$$\mathbf{y}_t = \begin{bmatrix} y_{t+1|t}, & \pi_{t+1|t} \end{bmatrix}' \qquad \mathbf{x}_t = \begin{bmatrix} y_{t|t-1}, & y_{t-1|t-2}, & r_{t|t-1}, & r_{t-1|t-2}, & \pi_{t|t-1}, & \pi_{t-1,3|t-2,4}, & 1 \end{bmatrix}'$$

$$\boldsymbol{\xi}_t = \begin{bmatrix} y^*_{t+1|t}, & y^*_{t|t-1}, & y^*_{t-1|t-2}, & g_{t|t-1} \end{bmatrix}'$$

$$\mathbf{H}' = \begin{bmatrix} 1 & -\hat{\alpha}_{y,1} & -\hat{\alpha}_{y,2} & \hat{\alpha}_g \\ 0 & -\hat{\beta}_y & 0 & 0 \end{bmatrix} \qquad \mathbf{A}' = \begin{bmatrix} \hat{\alpha}_{y,1} & \hat{\alpha}_{y,2} & \frac{\hat{\alpha}_r}{2} & \frac{\hat{\alpha}_r}{2} & 0 & 0 & \hat{\alpha}_0 \\ \hat{\beta}_y & 0 & 0 & 0 & \hat{\beta}_\pi & 1-\hat{\beta}_\pi & 0 \end{bmatrix}$$

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad \mathbf{Q} = \begin{bmatrix} \sigma^2_{y*} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & (\lambda_g \sigma_{y*})^2 \end{bmatrix} \qquad \mathbf{R}_t = \kappa_t^2 \begin{bmatrix} \sigma^2_{\tilde{y}} & 0 \\ 0 & \sigma^2_\pi \end{bmatrix}$$

Vector estimated by maximum likelihood:

$$\theta_2 = \begin{bmatrix} \hat{\alpha}_{y,1}, & \hat{\alpha}_{y,2}, & \hat{\alpha}_r, & \hat{\alpha}_0, & \hat{\alpha}_g, & \hat{\beta}_\pi, & \hat{\beta}_y, & \sigma_{\tilde{y}}, & \sigma_\pi, & \sigma_{y*} \end{bmatrix}$$

The signal-to-noise ratio $\lambda_z = |\hat{\alpha}_r|\,\sigma_z/\sigma_{\tilde{y}}$ is applied to a structural-break statistic in a regression of the gap-implied residuals on their determinants. The dispersion weighting enters this regression explicitly through weighted least squares with weights $w_t = 1/\kappa_t^2$,

$$\hat{\boldsymbol{\beta}} = \big(\mathbf{X}'\mathbf{W}\mathbf{X}\big)^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}, \qquad \mathbf{W} = \operatorname{diag}\big(1/\kappa_1^2, \ldots, 1/\kappa_T^2\big), \tag{23}$$

so that high-dispersion quarters receive proportionally less weight in the determination of λ_z.

Stage III Specification:

$$\mathbf{y}_t = \begin{bmatrix} y_{t+1|t}, & \pi_{t+1|t} \end{bmatrix}' \qquad \mathbf{x}_t = \begin{bmatrix} y_{t|t-1}, & y_{t-1|t-2}, & r_{t|t-1}, & r_{t-1|t-2}, & \pi_{t|t-1}, & \pi_{t-1,3|t-2,4} \end{bmatrix}'$$

$$\boldsymbol{\xi}_t = \begin{bmatrix} y^*_{t+1|t}, & y^*_{t|t-1}, & y^*_{t-1|t-2}, & g_{t|t-1}, & g_{t-1|t-2}, & z_{t|t-1}, & z_{t-1|t-2} \end{bmatrix}'$$

$$\mathbf{H}' = \begin{bmatrix} 1 & -\hat{\alpha}_{y,1} & -\hat{\alpha}_{y,2} & \frac{-\hat{\alpha}_r}{2} & \frac{-\hat{\alpha}_r}{2} & \frac{-\hat{\alpha}_r}{2} & \frac{-\hat{\alpha}_r}{2} \\ 0 & -\hat{\beta}_y & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{A}' = \begin{bmatrix} \hat{\alpha}_{y,1} & \hat{\alpha}_{y,2} & \frac{\hat{\alpha}_r}{2} & \frac{\hat{\alpha}_r}{2} & 0 & 0 \\ \hat{\beta}_y & 0 & 0 & 0 & \hat{\beta}_\pi & 1-\hat{\beta}_\pi \end{bmatrix}$$

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \qquad \mathbf{Q} = \begin{bmatrix} (1+\lambda_g^2)\sigma_{y*}^2 & 0 & 0 & (\lambda_g\sigma_{y*})^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (\lambda_g\sigma_{y*})^2 & 0 & 0 & (\lambda_g\sigma_{y*})^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \left(\frac{\lambda_z\sigma_{\tilde{y}}}{\hat{\alpha}_r}\right)^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{R}_t = \kappa_t^2 \begin{bmatrix} \sigma_{\tilde{y}}^2 & 0 \\ 0 & \sigma_{\pi}^2 \end{bmatrix}$$

Vector estimated by maximum likelihood:

$$\theta_3 = \begin{bmatrix} \hat{\alpha}_{y,1}, & \hat{\alpha}_{y,2}, & \hat{\alpha}_r, & \hat{\beta}_\pi, & \hat{\beta}_y, & \sigma_{\tilde{y}}, & \sigma_\pi, & \sigma_{y*} \end{bmatrix}$$

The filtered and smoothed states from this stage deliver the one-sided and two-sided estimates of the market-perceived natural rate of interest: $r^*_{t+1|t} = g_{t+1|t} + z_{t+1|t}$.

B Survey Forecast Data

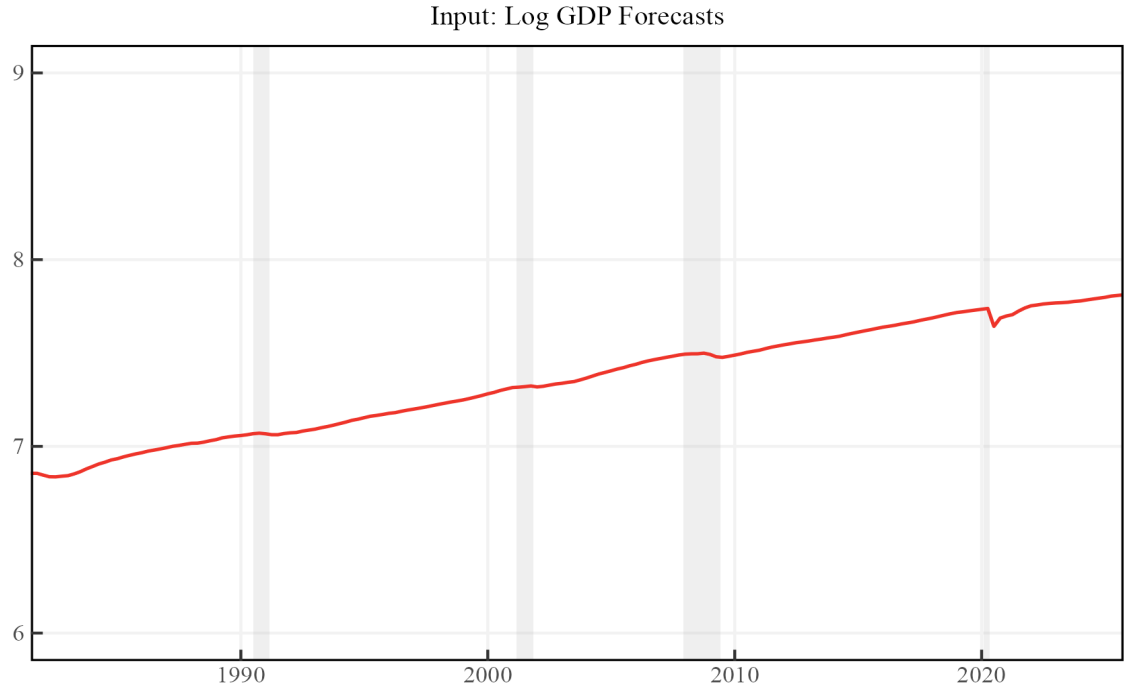


Figure B.1. Log GDP Forecasts, 1981:3-2025:4

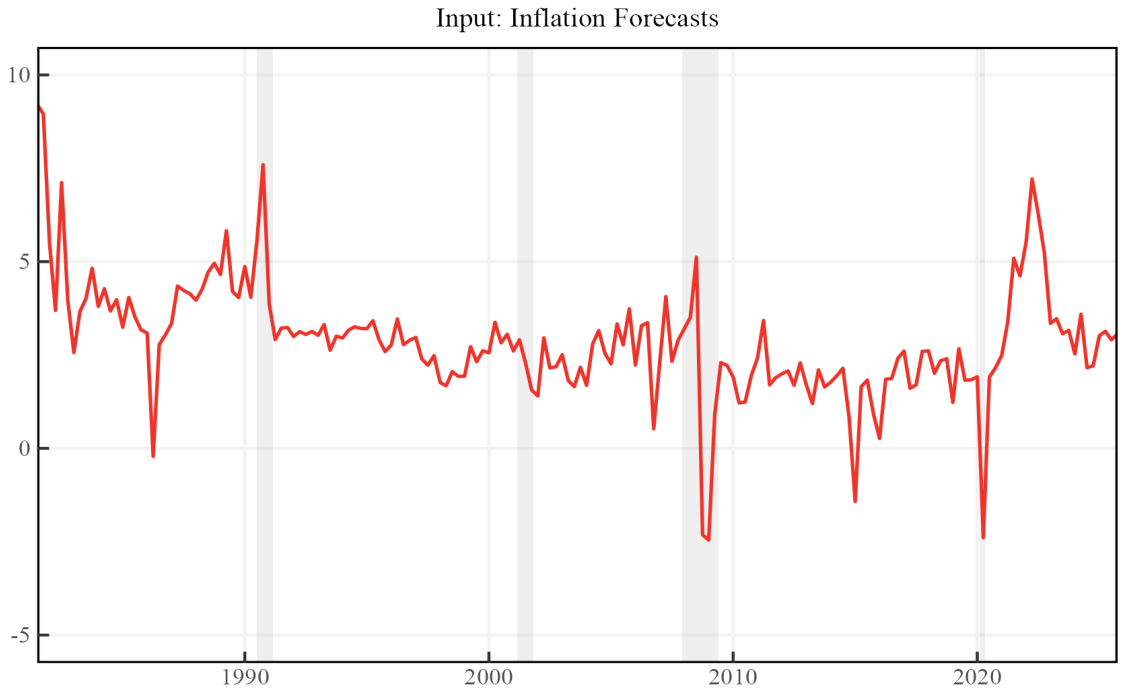


Figure B.2. Inflation Forecasts, 1981:3-2025:4

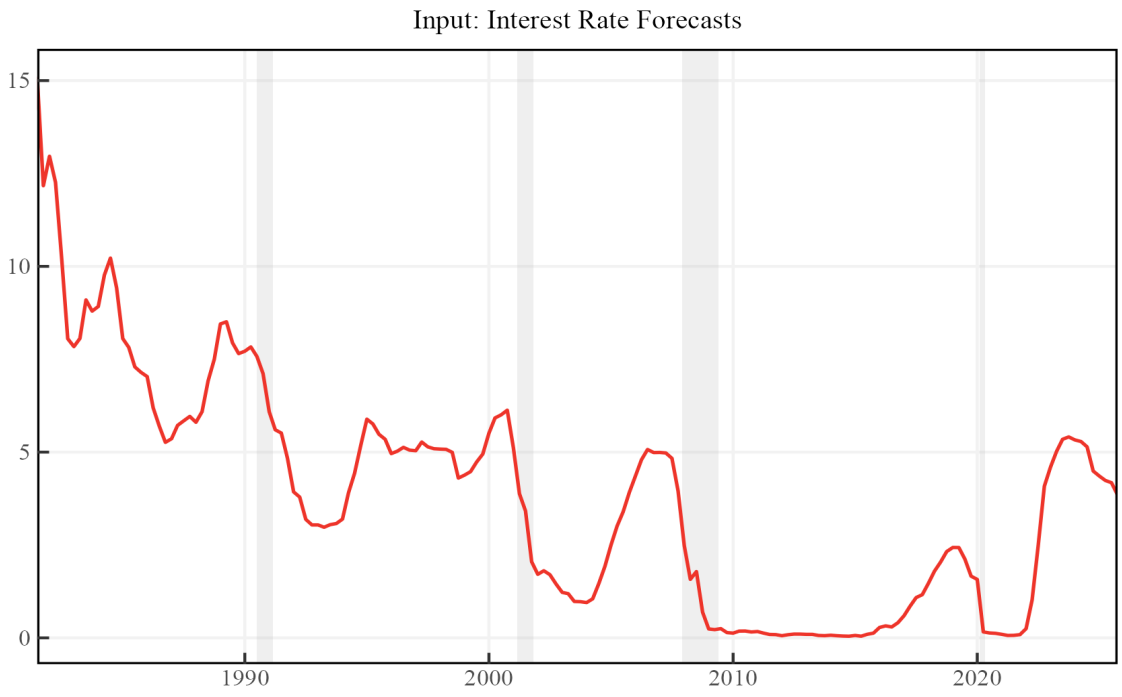


Figure B.3. Interest Rate Forecasts, 1981:1-2025:4

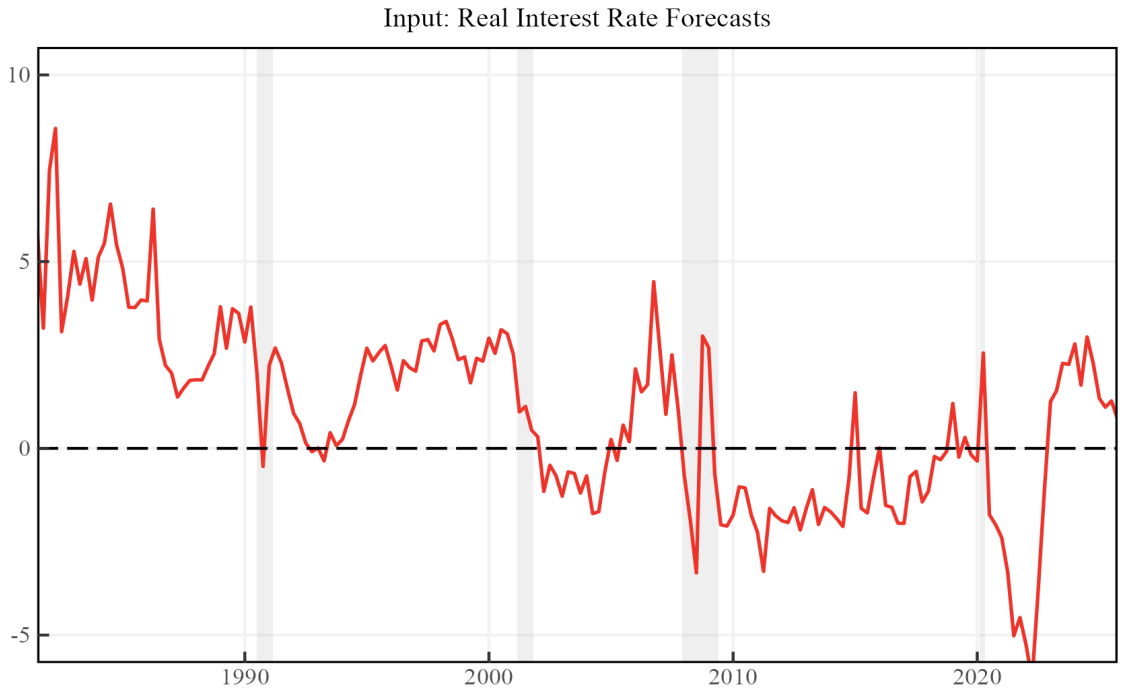


Figure B.4. Real Interest Rate Forecasts, 1981Q1-2025Q4

C Two-sided Estimates

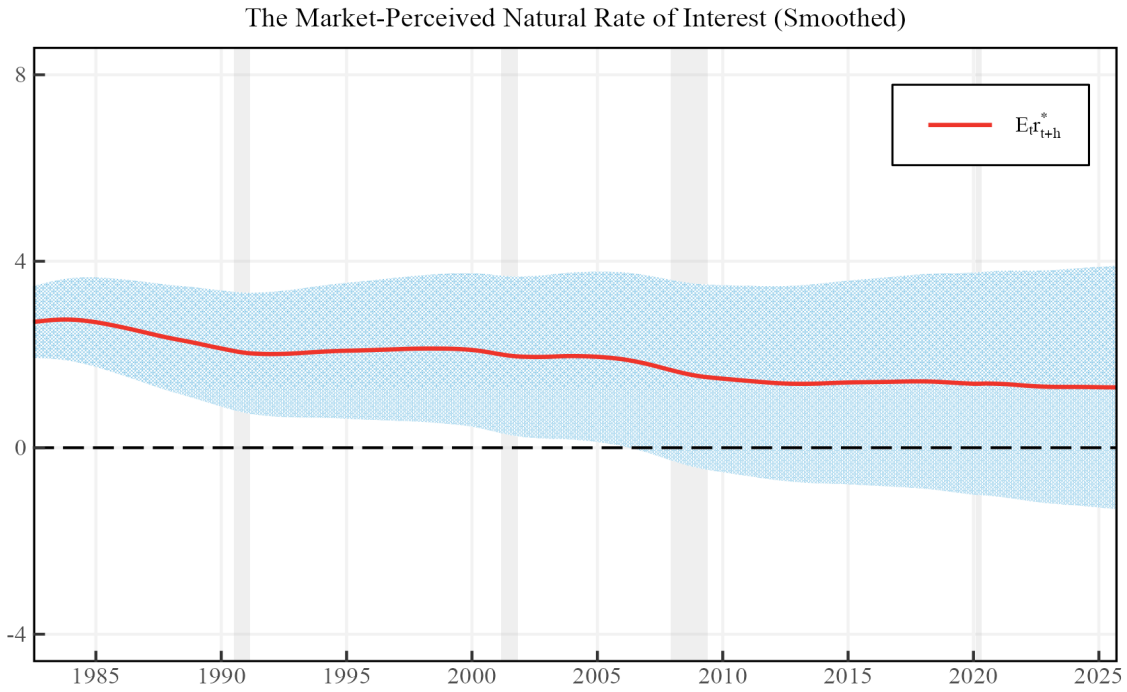


Figure C.1. The Market-Perceived Natural Rate of Interest (Smoothed)

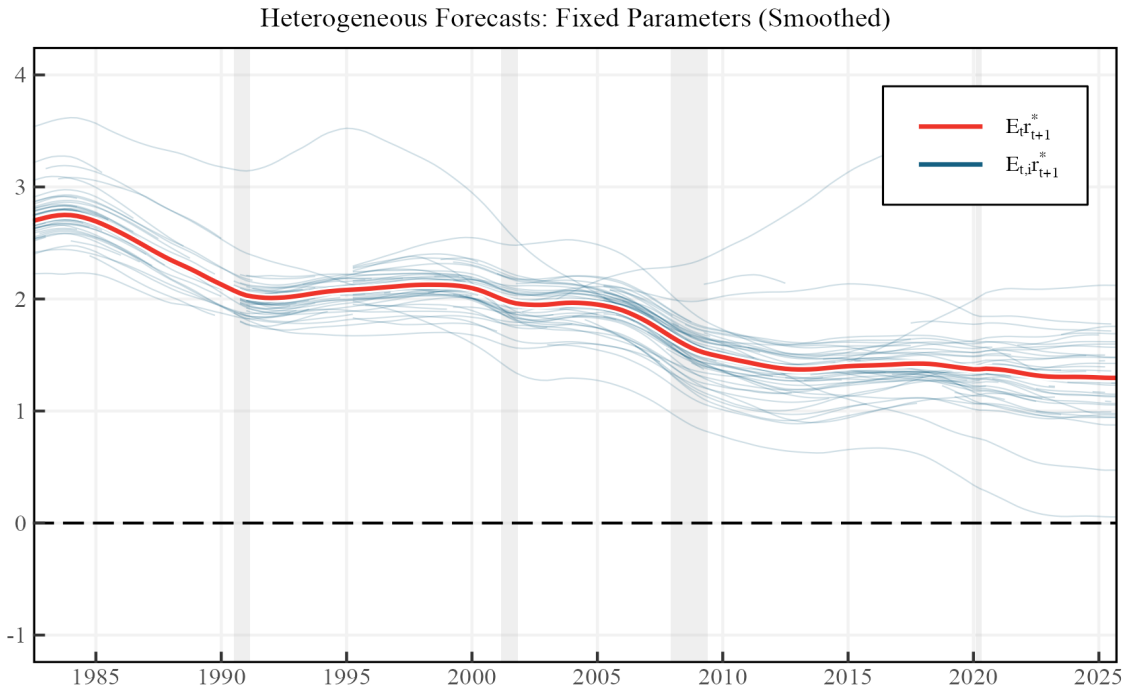


Figure C.2. Heterogenous Estimates: Fixed Parameters (Smoothed)

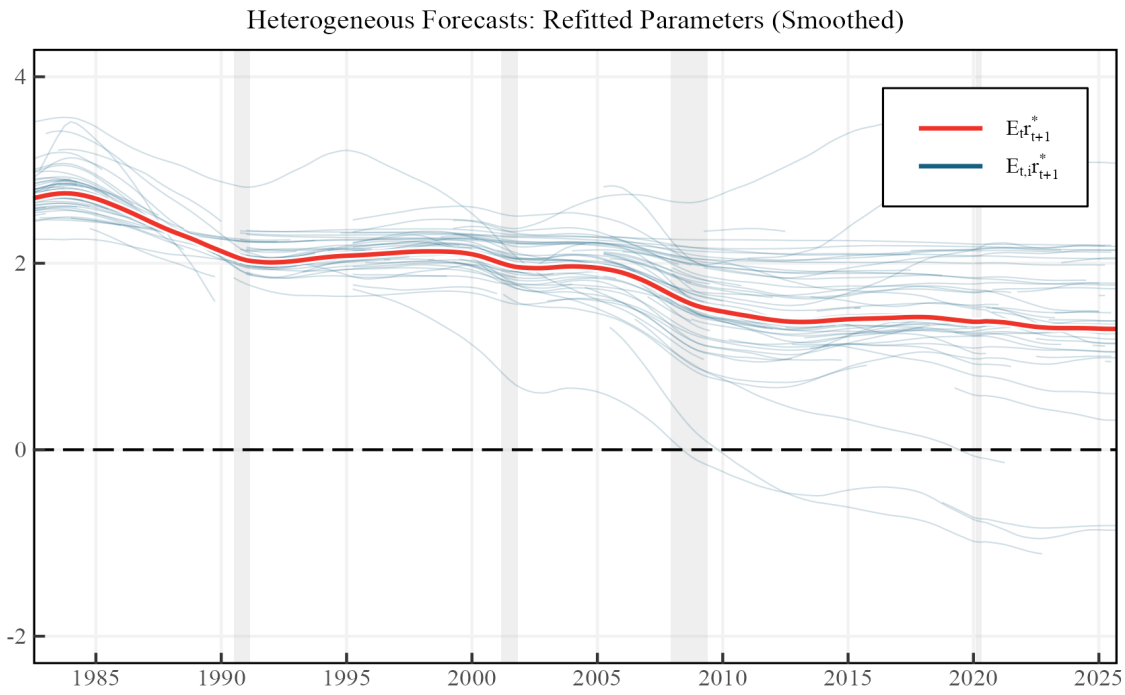


Figure C.3. Heterogenous Estimates: Refitted Parameters (Smoothed)

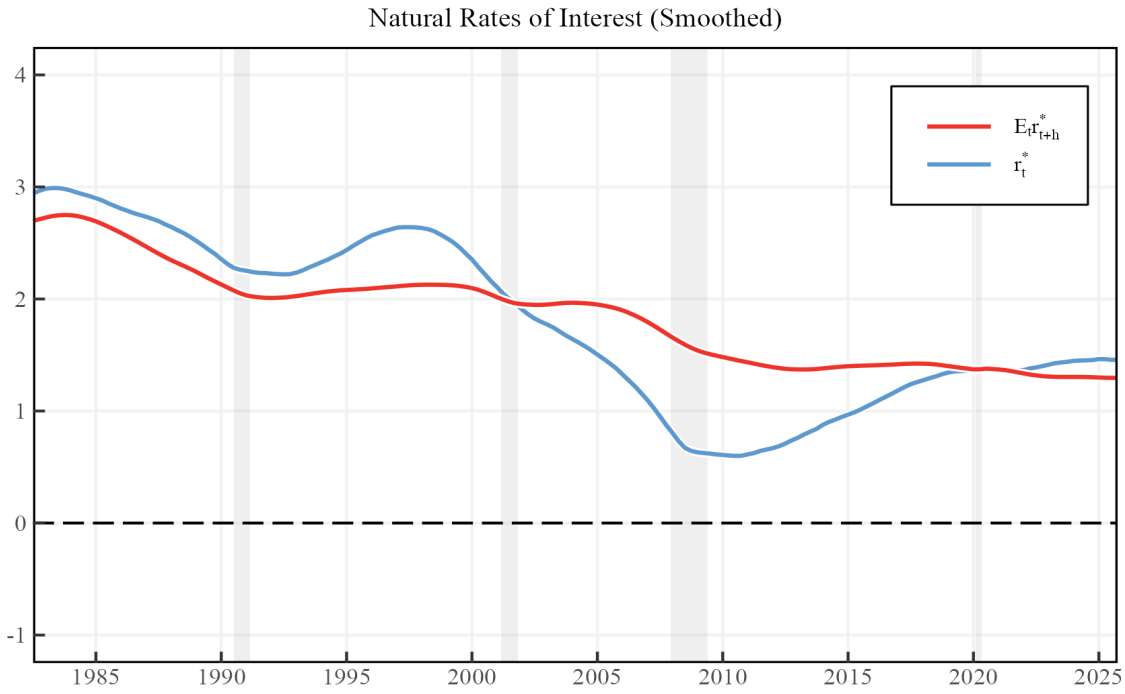


Figure C.4. Natural Rates of Interest (Smoothed)

D Kappa Normalisation

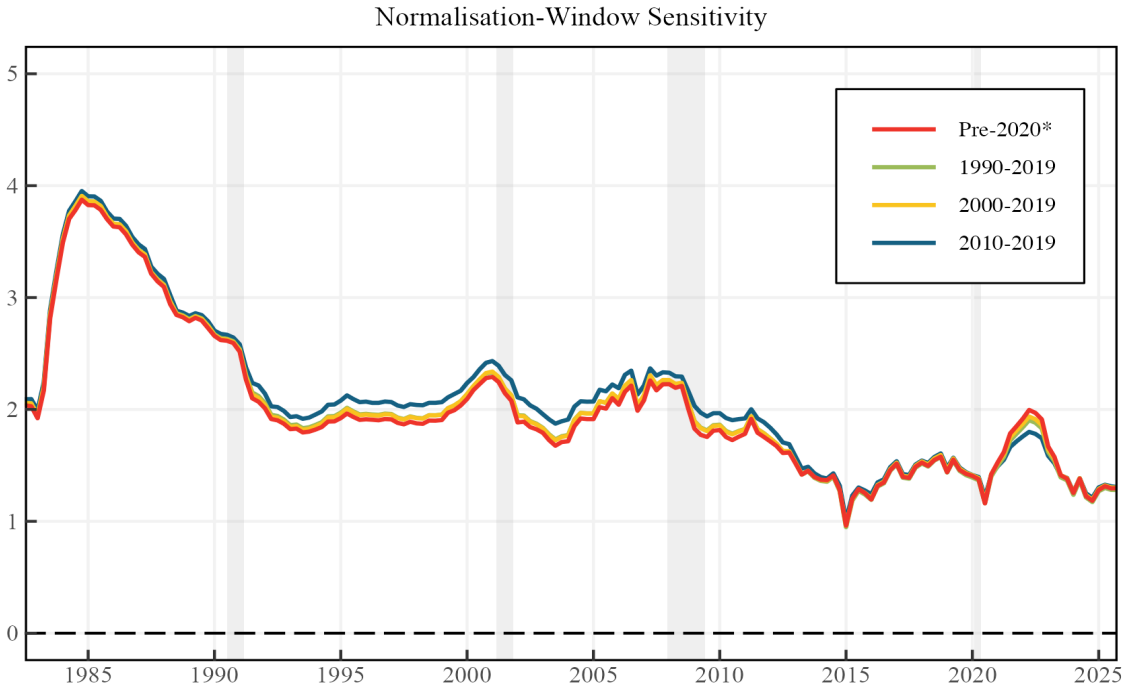


Figure D.1. Normalisation-Window Sensitivity

E Direct Survey Expectations

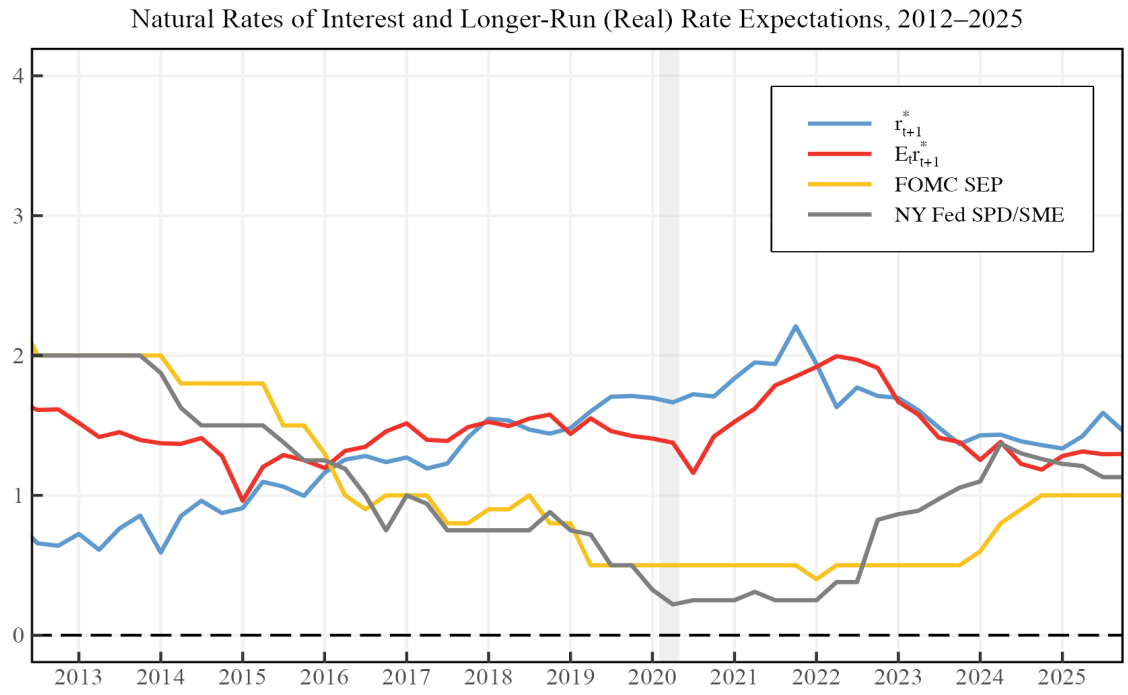


Figure E.1. Natural Rates of Interest and Longer-Run (Real) Rate Expectations, 2012-2025

F Variance Decomposition

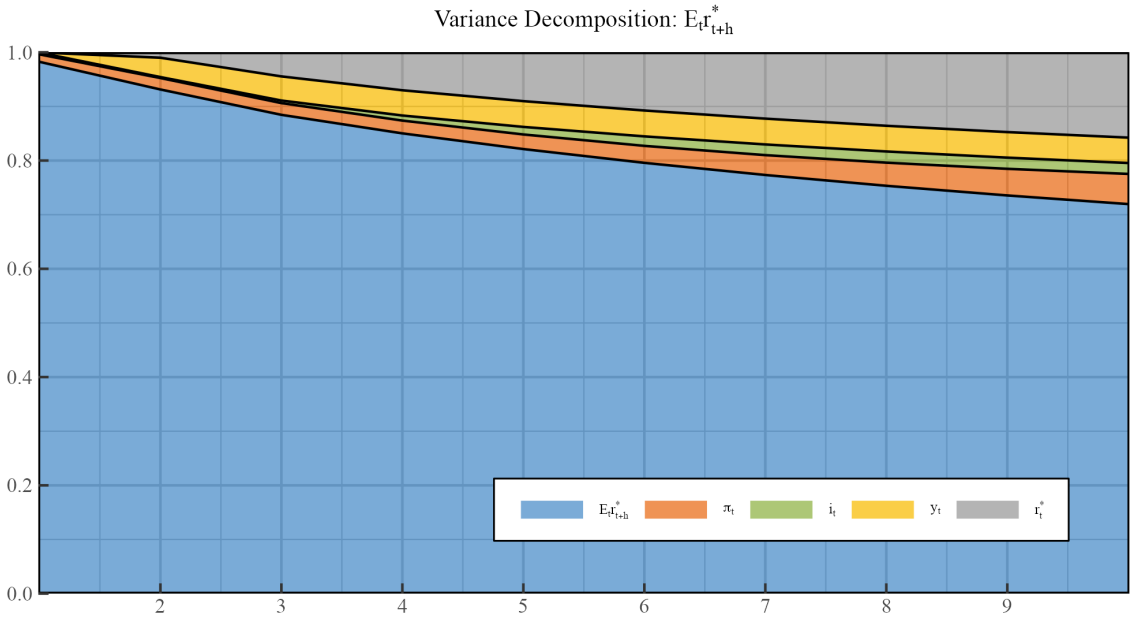


Figure F.1. Forecast Error Variance Decomposition

G Stationarity

Table G.1. Macro Drivers

ADF test (p-value)	y_t	π_t	i_t	r_t^*	$E_t r_{t+1}^*$
None	0.99	0.07	0.14	0.34	0.40
+ Drift	0.09	0.03	0.28	0.70	0.53
+ Trend + Drift	0.22	0.24	0.28	0.62	0.10

Note: n = 174 observations. Augmented Dickey-Fuller tests for the perceived natural rate, benchmark natural rate, short-term interest rate, consumer price inflation and the growth rate in gross domestic product.

Table G.2. Financial variables

ADF test (p-value)	$\tilde{r}_{t+1}^{*_e}$	ANFCI	SPREAD	CAPE	CREDIT
None	0.04	0.01	0.04	0.92	0.16
+ Drift	0.31	0.01	0.03	0.23	0.51
+ Trend + Drift	0.53	0.01	0.11	0.38	0.56

Note: n = 174 observations. Augmented Dickey-Fuller tests for the gap between the perceived and benchmark natural rate, Adjusted National Financial Conditions Index (ANFCI), ten-year two-year term spread (SPREAD), cyclically adjusted price-to-earnings ratio (CAPE), and the credit-to-GDP gap (CREDIT).